Modeling & Simulation of BSAR (Bi Static Synthetic Aperture Radar) system using Reflected BeiDou B1i Signal

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Dr. Muhammad Usman

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Abstract

BeiDou is the upcoming addition in the GNSS (Global Navigation Satellite Signals) community. Direct BeiDou signals are used for navigation and positioning purposes. This research intends to utilize the reflected signals since BeiDou provides an opportunity for passive radar applications. The BeiDou satellites, a modified BeiDou receiver and it's signal detection components (antennas), form a bi-static radar system. The Bi-static radar is used to detect the target or any movement or changes. This research reports the progress in Bi-static synthetic aperture radar using BeiDou as a transmitter of opportunity and a fixed ground-based receiver. The whole scenario has been simulated as closely as possible, and a match filter technique is used to improve the image resolution. The simulation has been carried out under different conditions and results have been described in a detailed manner, which shows that the BSAR using BeiDou as an illuminator or "transmitter of opportunity" can detect different targets with acceptable resolution.

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List of Abbreviation

AS	Anti-Spoofing
ADC	Analog to Digital Converter
BPSK	Binary Phase Shift Keying
DOD	Department of Defense
DSSS	Direct Sequence Spread Spectrum
FM	Frequency Modulation
FFT	Fast Fourier Transform
GO	Geometric Optics
GPS	Global Positioning System
I	In Phase
ID	Identification
IF	Intermediate Frequency
IFFT	Inverse Fast Fourier Transform
LHCP	Left Hand Circularly Polarized
LO	Local Oscillator
NAV	Navigation
OCS	Operational Control Segment
PRN	Pseudorandom Number
Q	Quadrature
RCS	Radar Cross-section
RF	Radio Frequency
RHCP	Right Hand Circularly Polarized
SAR	Synthetic Aperture Radar
SNR	Signal to Noise Ratio
SPS	Standard Positioning Services
SV	Space Vehicle

1. INTRODUCTION

The BeiDou system is a Chinese satellite-based navigation system. The system provides Global, Highly precise, reliable, and stable position and velocity information to its users. BDS development started in 1994 after that, the BeiDou constellation was deployed in three stages the installation of the demonstrated system (BDS-1) with two first-generation GEO satellites launched in 2000 and a third GEO satellite launched in 2003 to further enhance the system [1]. The BDS-second generation COMPASS/BDS-2 operation started with the successful launch of one GEO experimental satellite and one Medium Earth Orbit (MEO) satellite and has achieved full operational capability in 2014 providing the position and navigation service to their user over the whole Asian-Pacific region with the 14 satellites, 5 GEO, 4 MEO, and 5 highly inclined GEO satellite. The BDS-3 system consists of three GEO three IGSO and 24 MEO satellites and some additional satellites in orbit for backup [2]. The GEO satellites are located at 80^0 E, 110^0 E, and 140^0 E respectively with an altitude of 35786km. the BDS IGSO satellites revolve in plane at altitude of 35,786km, 55^0degrees inclined with reference to equatorial plane and the MEO satellites orbit in three orbital planes, 9 satellites in each at altitude of 21,528km with 55^0 degree inclination to reference equatorial plane[3]

The reflected GNSS signal provides an opportunity for passive radar system applications [4]. The primary cause of developing a GNSS system is accurate navigation but the reflected GNSS signal has great potential and has been utilized for remote sensing with great attention in the last two decades [5].

BeiDou based Bi-static radar system utilized BDS as a transmitter of opportunity a modified receiver and it can also be used for remote sensing [6]. The separation of transmitter and receiver is called Baseline in Bi-static radar and making an angle called the bi-static angle. The bi-static radar Scenario based on BDS is depicted in Fig 1.1

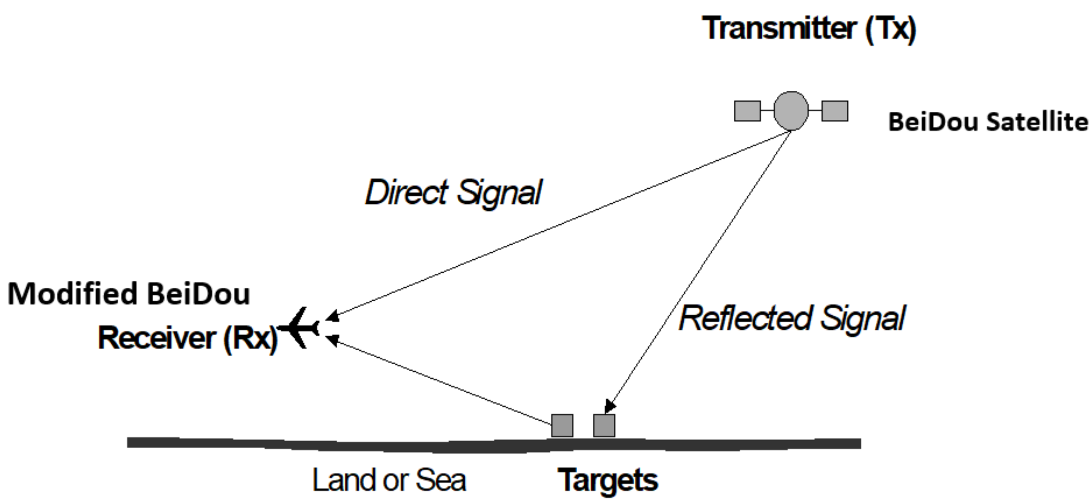

Fig 1.1 Schematic representation of the Scenario

BDS signals are utilized in Bi-static radar as the “transmitter of opportunity” to illuminate the object and a modified BDS receiver is to process these reflected signals. This type of radar that uses a “transmitter of opportunity” is said passive bi-static radar [4].

The passive bi-static radar advantage is that it requires no dedicated transmitter. BDS-based bi-static radar consumes less power as compared to a heavily dedicated transmitter and has also the ability to detect stealthy targets as bi-static radar geometry is different from mono-static Radar [6].

As mentioned before direct BDS signal is utilized for position and velocity measurement, however direct BDS signals are also reflected from a static or moving object as they strike off from the reflecting surface. That reflected signals have very low power and SNR because of low signal power and after reflection from the object, power is further reduced, which makes them difficult to detect. After implementing suitable signal processing techniques reflected signals can be used in varied remote sensing applications [7]. After reflection from the surface, signals have valuable information about the reflecting surface, and the signals can also be utilized for generating an image of the target in the area of interest. The BDS reflect signals can have numerous uses, such as measurement of soil moisture content, passive microwave imaging, wind speed measurement, and ocean remote sensing etc. [8], [9][10].

The GNSS-SAR based target detection and imaging eliminate the need for a dedicated transmitter. BDS navigation infrastructure already being used and be utilized for target detection and imaging. BDS has an expensive infrastructure already installed and maintained provide to be a huge edge for the user. The BDS is available all day long and covers the complete surface of the earth which is a huge advantage. For more signal power and visibility (specular reflections), an optimum geometry of the BDS satellite has to be selected [2].

BDS navigation system development increases signal availability and its signals are freely available for public use. Due to this BDS based remote sensing is very attractive for research and there are ample opportunities.

The BDS transmitted radio navigation signals are right hand circularly polarized (RHCP) and after striking the earth surface or any object it reflects, and its polarization is changed. These signals may become left hand circularly polarized (LHCP). For acquiring and detecting reflected signals the direct signals from a specific satellite are received and lock to use for reference[11]. The receiver available in the market is not able to receive the reflected signals because they have only RHCP antenna which can only receive direct signals. The BDS signals received and acquired data in digital form and saved for offline processes to perform remote sensing and image the area of interest. A high gain directional antenna is used for receiving BDS data in form of reflected signals. The circuit that acquires data has a suitable intermediate frequency (IF) and sampling frequency. An extensive problem is the reception of weak signals owing to their appalling SNR and constant data rate of acquiring the BeiDou satellite system IF by appropriate means this received data is transferred to the processing system[12].

The direct and reflected BDS signals are manipulated to generate the image of the area of interest. The (Synthetic Aperture Radar) SAR principle is used. The reflected BDS signals from objects on earth will be acquired by suitable hardware, as mention above, this hardware fixed on a moving object (an Arial platform or UAV) for synthetic aperture effect and provide the requisite change in geometry as shown in Fig 1.1 to the reconstruction of an image from reflected BDS signals based on bi-static radar configuration, the range resolution is a very important parameter. On the same bearing, the Radar system can detect two or more targets at different ranges[11]. The range resolution is given by.

$$\delta_r = \frac{c}{2B\cos(\frac{\beta}{2})} \tag{1.1}$$

Where B is the bandwidth of the transmitted signal and β is the bistatic angle. Range resolution depends on the bandwidth of signals and the bi-static angle. For better resolution improve the bandwidth of the signal. In our case, we use transmitter opportunity and cannot improve the bandwidth because we have no control over the BDS transmitter, the parameters which can be modified is the bi-static angle. Maximum resolution can obtain at zero angle at which bi-static radar becomes mono-static. BDS signal is transmitted at higher bandwidth therefore significantly improve the image range resolution**.**

BDS signals are presently transmitted at B1 frequency with ranging code. The ranging code is unique for each satellite and freely available for all civilian users. this new signal will be more accurate and compatible for remote sensing with compare of GPS 'L1' signals. On 3 August 2020 in a news conference at the state council information Office and announced that BDS-3 was formally commissioned[13]

1.1 Remote Sensing advantages and disadvantages

The expensive BDS infrastructure maintain for navigation purposes has the attraction for the user to take advantage for target detection, imaging and there are more simultaneous opportunities of measurement one for each BDS satellite in view. Cost-effectiveness is the biggest advantage. The BDS receiver, comparable in size and complexity to a notebook computer, can be built for a fraction of the cost of traditional radar, space-borne equipment, and other sensors.

The reflected signal low power level is the primary disadvantage as compared to ordinary radar. Various techniques are used to address this problem and partially overcome this problem by utilizing multiple receiving antennas with different gain, polarization sensitivities, and longer observation time. This issue is discussed later in detail and some alternate methods have also been suggested in this regard.

1.2 Organization of Report

The report can be divided into 6 chapters.

It is necessary to familiar with the BDS system fundamental to understand the BDS reflected signal. In the second chapter, the report the BDS system is elaborate deeply, depicting its working principle and concepts that are important to understand this vital technology tool. This chapter starts with the basic introduction with the GNSS system presently operational in the world, then later on introduction with the three segments of the BDS system and then describe the working principle of the system and the characteristic of BDS signal structure. Other useful properties as modulation, auto-correlation, noise, interference, and source of error are also mentioned. The SAR (Synthetic Aperture Radar) theory is also mentioned in brief and matched filter and the FM chirp signal is also with short write up. These concepts presented in the second chapter is important to understand the fourth simulation chapter.

The third chapter of the report describes the signal acquisition process of the receiver and explain the basic working principle of a generic BDS Receiver and its basic block diagram. The chapter also explains the proposed remote sensing BDS receiver with its block diagram.

Tn the fourth chapter of the report the simulation work can be discussed. This work was carried out on the MATLAB version 2014a. The simulation process can be divided into two categories. The first type is the simulation of the real world i.e. the processes that occur in BDS satellite and in space before to reach the signal to receiver and simulation of the BDS receiver. The processes that simulated during this research work is stated as follows. Satellite orbit and coordinate system, code offset, code generation, Noise, attenuation, and image construction. The written in MATLAB is explain and elaborated in word.

In the fifth chapter of the report discussed the results and concluded the simulation process. Different images show under the different conditions when the simulation was carried out by varying the parameters like sampling time, noise, and attenuation. Simulation findings and shortcomings are analyzed.

The sixth chapter is the final chapter of the report and summarized the project and furnishes the conclusion and recommendation. Finally, suggestions for future work and advanced research that can be undertaken in this field are proposed. The copy of the MATLAB code is also attached as appendices.

2. BRIEF OVERVIEW OF GNSS AND SAR

2.1 Global Navigation Satellite Systems (GNSS)

Global Navigation Satellite Systems (GNSS) constellation is consisting of satellites revolving around the earth to provide continuous position and timing information to their user to have appropriate equipment everywhere on the earth or near the earth surface. GNSS comprises all the satellites-based navigation system such as USA's Global Positioning System (GPS), Russian GLONASS, Europe's Galileo and China BeiDou [14][15]. The comprehensive detail of these GNSS system is given in below Table 2.1

Table 2. 1

Comprehensive Details of GNSS Systems

Parameters	GPS	GLONASS	Galileo	BeiDou
No. of Satellite	Operational 24 spares7	Operational 21 spares 3	Operational 24 spares 3	27MEO+5 GEO+3IGSO
Orbital Planes	6	3	3	6
Inclination	55	64.8	56	55
Orbital Height	22,200km	19,100km	23,222km	21,150km
Orbital Period	11hrs 58min	11hrs 15min	14hrs 04min	12hrs 37min
Ref. Time	GPST	GLONASST	GST	BDT
Ref. Frame	WGS-84	PZ90	GTRF	CGCS 2000

Where

GPST, GLONASST, GST, and BDT stands for GPS, GLONASS, Galileo and BeiDou time, respectively. These systems detailed discussed.

2.1.1 The Global Positioning System (GPS)

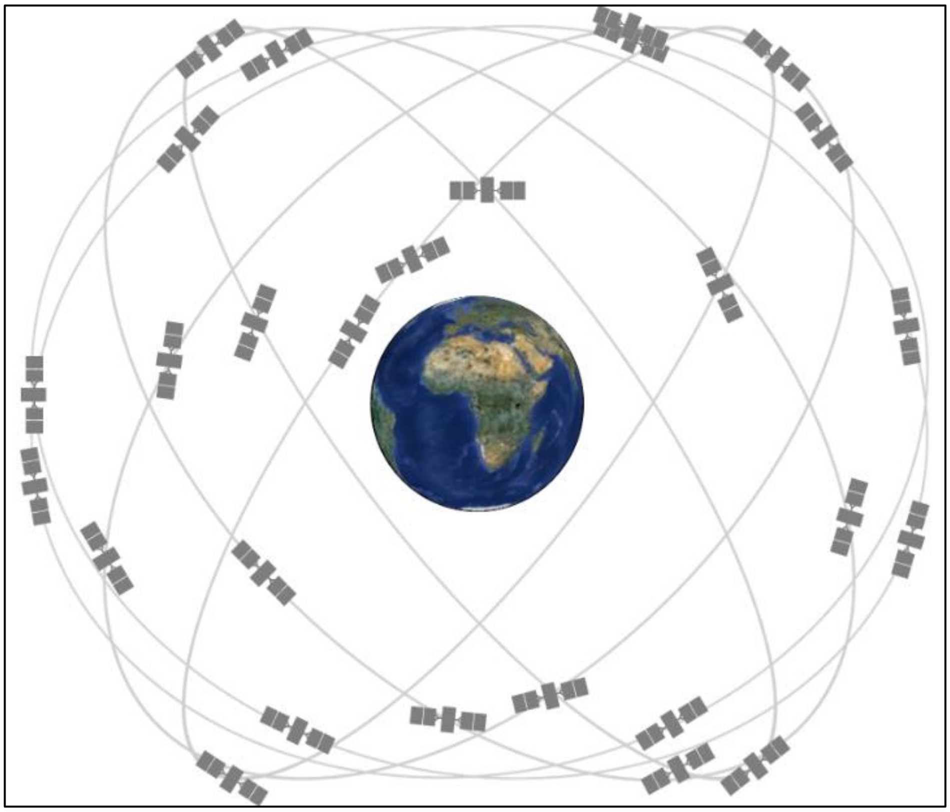

Fig 2. 1 GPS current constellation

The GPS is a US navigation satellite System, and it is the first GNSS system in the world. Working on the system development started in 1983, and in 1995 it achieved full operational capability. The system provides information about the position, velocity, and time to its user in real-time over the Globe. GPS is the most mature satellite system in the world. The system consists of 30 satellite which orbits in MEO orbit at an altitude of

20200km and the inclination with equator is 55^0 degrees. These satellites are arranged in 6 orbital planes 4 satellites in each and 1 spare satellite for backup. These orbital planes are separated by 60^0 right ascensions of the ascending node (angle along the equator from a reference point to the intersection of the orbit). the orbital structure will ensure that every point on earth at least four satellites in view. Orbit eccentricity is 0.02 nearly circular and the semi-major axis is 26560km and 11h58m2s is a nominal period, repeating the ground track in each sidereal day[16]. GPS current constellation is shown in Fig 2.1.

2.1.2 GLONASS

The GLObal'naya NAvigationnaya Sputnikovaya Sistema (GLONASS) is the second satellite system. The GLONASS is the satellite-based navigation program, initiated in mid 1970s by the Soviet Union, system is for both civilian and military purposes similar to GPS[17][18].the GOLNASS first satellite was launched in 1982[19] and completed the constellation of four satellites in 1984. The operation of the satellites launched is continued till the mid of 1990s. In February it announced that the system is fully operational when the system 24 satellites in space[20]. The Russian GHLONASS developed parallel to the GPS, but with a few year's lag. However due to economic issues and other issues , the number of satellites decreased from 24 to 6 in 2001 and after 2001 the Russian government decided to recover the constellation and modernized the system. In 2011 the system restores its full operational capability[21]. The GLONASS constellation is deployed in three orbital planes. GLONASS constellation had 24 MEO satellites and 6 spares satellites, 8 satellites in each plane, and equally spaced. The orbit inclination is 64.8^0 and roughly circular, at the altitude of 19100km had a period of

11h15m44s, repeating his geometry every eight sidereal days[22]. The GOLNASS satellites constellation shown in the Fig 2.2

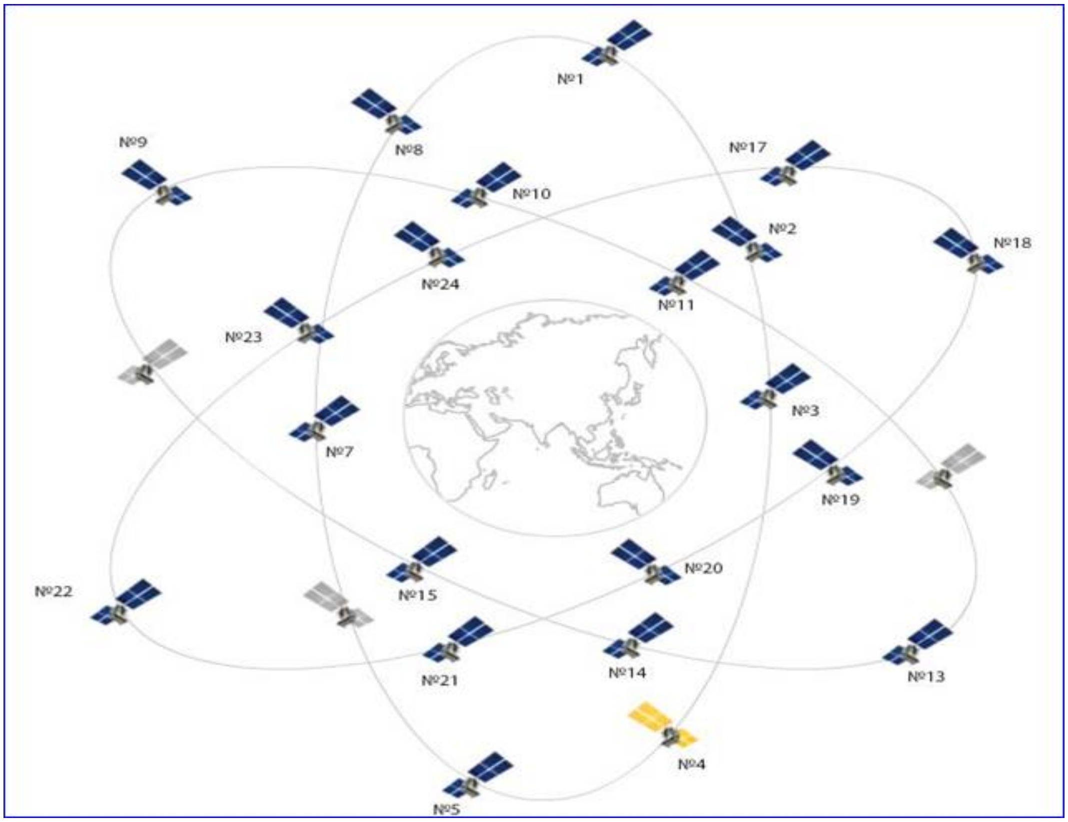

Fig 2. 2 GLONASS Constellation

2.1.3 Galileo

The European Geostationary Navigation Overlay System or EGNOS is the result of Europe's GNSS policy in early 1990s. the system is a space base augmentation system (SBAS). Later on the European Commission (EC) and European Space Agency (ESA) is extended the project to European GNSS System known as Galileo in 1999[23], however the launched of first tow satellites GIOVE-A and GIOVE-B in space was made possible in 2005 and in 2008 respectively[24]. Here the Galileo In Orbit Validation phase Element is abbreviated as GIOVE[25].

In 2013 the Galileo first phase, the ground segment is completed after five year with the successful launch of 4 IOV satellites. In December 2016 European commission declared the Galileo initial services operational based on these satellites' services. The Galileo system consist of 30 satellite and the constellation deployed in three orbital planes 9 satellite in each plane and 1 spare satellite in each plane for backup. These three orbits are circular and in Medium Earth Orbit at an altitude of 23222km above the earth. The orbit inclination is 56^0 with respect to the equator and has an orbital period 14h4m45s, satellite repeat the geometry in every 10 sidereal days. The constellation structure will ensure at an elevation of 10^0 above the horizon minimum six satellite in view at any point on earth[1]. It is expected that the Galileo system can achieved its fully operational capability(FOC) in 2020 under European Space Agency. Some basic information about Galileo FOC program is given in Table.

Table 2. 2

Galileo Parameters

<table>
<tr><th>Parameter</th><th colspan="4">Value</th></tr>
<tr><td>Satellites(FOC)</td><td colspan="4">27operational+3 ACTIVE Spares</td></tr>
<tr><td>Orbital planes</td><td colspan="4">3</td></tr>
<tr><td>Slots per Orbital Planes</td><td colspan="4">8</td></tr>
<tr><td>Orbital inclination</td><td colspan="4">56^0</td></tr>
<tr><td>Orbital Radius</td><td colspan="4">23,222km</td></tr>
<tr><td>Repeat cycle</td><td colspan="4">17Orbites/10 Sidereal Days</td></tr>
<tr><td rowspan="2">Galileo Signals + Services Type + Frequencies(MHz) +</td><td>E1</td><td>Open</td><td>1575.42</td><td>CBOC 1/11</td></tr>
<tr><td>E5a</td><td>Open</td><td>1176.45</td><td>BPSK-R</td></tr>
</table>

Modulation type	E5b	Open	1207.14	BPSK-R
	E6	Commercial	1278.75	BPSK-R

2.1.4 The Chinese Navigation system BeiDou(BDS)

BeiDou is the first operational regional satnav system, and the first satnav system with plans to operates as both regional and global system. The BDS system is a recently developed Chinese global navigation and positioning system. The system is intended for dual use designed and operated to provide services open and authorized to its users[26]. BDS phase-1 was developed in the 1990s. The BeiDou Phase 1 initial segment was constructed in 1990 and the tow GEO satellite was launched in 2000 and declared operational in 2001.

Later the system development is extended to regional and then global, the system deployment has three phases[27]. Which are explained below.

2.1.4.1 BDS-1

BDS phase-1 was developed in the 1990s. The BeiDou Phase 1 initial segment was constructed in 1990 and the tow GEO satellite was launched in 2000 and declared operational in 2001. After the completion of the development phase, the BDS-1 first GEO satellite launched in Oct 2000 and second in Dec 2000, and the third satellite is launched in March 2003.

After the launch in late 2000, the system went through an on-orbital validation. The satellite that launched in Oct and Dec 2000 placed on 80^0 E and 140^0 E longitude on the Geostationary belt and the satellite launched in March 3003 is placed at 110^0E longitude [16].

2.1.4.2 BDS-2

The operation for BDS-2 was started with the successful launch of one MEO and one GEO experimental satellite in 2007 and 2009 respectively by the end of 2012 the BDS-2 the regional system get its operational capability and provide the positioning and navigation services to their user over the whole Asian-pacific region. With a constellation of five satellites in GEO orbit and five satellites in IGSO orbit and four in MEO orbit.

2.1.4.3 BDS-3

The BDS-3 mission started with the launch of the BDS-3 experimental satellite on March 30, 2015[28]. The space segment of BDS-3 is consisting of three GEO, three IGSO, and 24 MEO satellite accompanied by some additional satellite for backup[29].

2.1.5 Space Segment

The BeiDou space segment consists of 35 Satellite 27 in semi-synchronous orbits 3 in GEO and 5 in IGSO orbits. Semi-synchronous orbit satellites arranged in 3 orbital planes 9 in each. The satellite orbit altitude is 21528 Km above the earth surface. The orbital plans incline with the equator at 55 degree. GEO satellites orbits at an altitude of 35,786Km and located at 80E 110.5E and 140E, respectively. IGSO satellite orbiting at an altitude of 35,786Km with reference to the equatorial plane. The system gets its full operational capability in 2018 and some the satellite kept spare for Backup[3] and become operational in2020 August[30].

2.1.6 Control segment

Master control station (MCS), Monitor Station (MS), upload station, and Ground antennas are the necessary part of the ground control segment. BeiDou has one MCS located in Beijing. The MCS is responsible to monitor and manage the constellation it is the central processing facility for the control segment. The MCS is responsible to control satellite station-keeping maneuvers, continuously updated the navigation data message transmitted by the satellite, reconfiguration of redundant satellite equipment, and various other health monitoring and health activities. BDS monitor station is responsible for passively tracks all BDS satellite in view and collect ranging data from each satellite. MCS estimated and predicted ephemeris and clock parameters of satellites. Upload station to be used to upload ephemeris and clock data intend to retransmission in the navigation message to BDS receivers[31].

2.1.7 User segment

In the user segment, a user has the necessary equipment to received and process the received signal specially designed to received and decodes the BDS satellite signals.

2.2 Principle of working

The BDS each satellite carries several high accuracy atomic clocks and codes transmitted at a precisely known time. The BDS used time of arrival TOA ranging concept to determine the user position. The time is calculated from the signal is transmitted by SV at a known location to reach a user's receiver. The time difference is calculated to signal transmitted from SVs at a known location and reception at the receiver, the receiver can calculate its position. From the trilateration method receiver can calculate its position, it is a traditional, simplest, and most accurate method of locating an unknown position.

It is necessary to choose a reference coordinate system to solve the satellite navigation problem. The system in which both the satellite and the receiver are represented. CGCS 2000 system is used to measure and determine the orbit of BDS satellite in this system the origin is at the center of the mass of the earth[31].

2.3 BDS orbit

BDS satellite orbit accurate information is necessary to calculate the receiver position. For this, it's necessary to understand the characteristic of BDS orbits. The satellite orbit parameter is represented in various methods. The orbital parameter, time of their applicability, and their characterization change with time include in the ephemeris message. Based on this information the receiver can calculate the corrected integral of motion for SV to solve the navigation problems. In Table 2.1 the orbital elements are defining which is used in the algorithm by which a receiver calculates the position vector of the satellite (Xs, Ys, Zs) in the CGCS2000 coordinate system. These parameters are further discussed in the simulation section. In the table below the symbol and their description is given which define the orbit orientation in the CGSC2000 coordinate system.

Table 2. 3

BeiDou Ephemeris Data Definitions

SNR	Symbol	Description
1	t_{oe}	Reference time of ephemeris
2	$\sqrt{a}$	Square root of semi-major axis
3	E	Eccentricity
4	I_0	Inclination angle (at time t_{oe})
5	Ω_o	Longitude of the ascending node (at weekly epoch)
6	Ω	Argument of perigee (at time t_{oe})
7	M_0	Mean anomaly (at time t_{oe})
8	$di/_{dt}$	Rate change of inclination angle
9	Ω	Rate change of longitude of the ascending node
10	Δn	Mean motion correction
11	C_{uc}	Amplitude of cosine correction to argument of latitude
12	C_{us}	Amplitude of sine correction to argument of latitude
13	C_{rc}	Amplitude of cosine correction to orbital radius
14	C_{rs}	Amplitude of sine correction to orbital radius
15	C_{ic}	Amplitude of cosine correction to inclination angle
16	C_{is}	Amplitude of sine correction to inclination angle

In case of BeiDou constellation, the orbit is nearly circular with eccentricity no greater than 0.02, and approximately semi-major axis is 6378km. the remaining orbital parameters vary b/w satellites uniform coverage to the entire earth provide by the constellation.

BeiDou satellite transmitted almanac and ephemeris data (in the receiver section more explained) also included osculating Keplerian orbital elements. The Keplerian orbital element allows the user to estimate the Keplerian element accurately during the period of the time between updates of the satellite ephemeris message. For the necessary computation, it’s important to know the rotation rate of the earth. According to CGCS2000, the rotation rate is $\Omega_{e} = 7.2921150 \times 10^{-5}$rad/s[3].

2.4 BDS services

BDS is intended for dual use designed and operated to provides services both to civilian users and to authorized. The BDS system provides open and authorized services.

2.4.1 Open service

The open service of BDS is like other GNSS systems, BDS provide their service free of cost to user’s having the necessary equipment. The user having appropriate equipment can able to compute their navigation solution using the same principle as for the other GNSS systems and the system provide position accuracy up to 10 meters and time accuracy up to 50ns and velocity accuracy up to 0.2meter per second[1].

2.4.2 Authorized services

This service will ensure very reliable use providing safer positioning, velocity, and timing services, as well as system information , authorized users. B1Q, B2Q and B3 are only for authorized users

2.4.2.1 BDS-3 planed services

The BDS-3 provide the fundamental services to their user’s like other GNSS. Some special services which are exited in BDS-1 and BDS-2 are still managed like

SMCS regional area and satellite-based augmentation services. In BDS-3 additional services will be added like PPP, Global short message communication, space environment service. some of these services will be embedded in the GEO satellite and Some in the MEO satellite.

2.4.2.2 Precise point positioning

BDS-3 will provide PPP service to the user with three GEO satellites via a B2b frequency signal. The error correction parameter like orbit error, satellite clock error, and inter code bias will be broadcast to users of China and the surrounding area via B2b signal. User's in the service area can improve their accuracy by using these broadcast correction parameters up to decimeter level kinematic positioning and centimeter-level static positioning.

2.4.2.3 Short message communication services

BDS can provide this service regional and as well as globally. The regional short message communication service (RSMCS) is provided via three GEO satellites with a capacity of 1000 Chinese characters at one time. The Global short message communication service will be provided via 27 MEO satellites.

2.4.2.4 Search and rescue services

BDS-3 coordinate with others SAR constellation like COMPAS, SARSAT to provide free SAR service to global maritime, aviation, and land users[32].

2.5 BeiDou Satellite operations

The BDS satellite navigation system all satellite transmit right hand circular polarized signal centered at three L band carrier frequency first one called B1 at 1575.420

MHz, the second one called B2 at 1191.795 MHz, and the third is called B3 at 1268.520 MHz All signal is the sum of the channel I and Q phase which are the in-phase quadrature of each other. The ranging code and navigation message modulated on these carrier frequencies.

The satellite navigation signal is composed of the carrier frequency, ranging code, and navigation data. The CDMA technique is used to multiplexing the signal as used for GPS and Galileo. The Beidou signals their structure and support services are summarized in Table 2.2.

Table 2. 4

Signal Characteristic of the BeiDou satellite system

Band	Carrier frequency/MHz	PRN rate (Mcps)	Chip rate/Mcps	Bandwidth (MHz)	Modulation type	Service type
B1	1561.098	B1I	2.046	4.092	QPSK(2)	Open
		B1Q				authorized
B2	1207.14	B2I	2.046	24	BPSK(2)	Open
		B2Q	10.23		BPSK(10)	Authorized
B3	1268.52	B3	10.23	24	QPSK(10)	authorized

2.6 BDS Signals

The BDS-3 system transmits signals on three frequencies B1 1575.420MHz, B2 1191.795MHz, and B3 1268.520MHz. BDS-2 signal B1I and B3I are still maintained, and BDS-3 also transmitted the signal for interoperability B1C. The BDS-3 MEO satellites will be transmitted the signal B3I, B1C, B2a, and B2b for the global users. The regional user's, in the Asia-Pacific region, B2a is used for SBAS, and B2b is used for PPP.

2.6.1 Nemon halfmen code

The NH code is the composition of 2046 bit pseudorandom noise (PRN) code having a clock rate of 2.023 MHz which repeats every 1 milliseconds. The BeiDou each satellite has a different PRN code to other satellites and these code sequences is selected from a set of two codes combination called gold codes. The Gold code is selected for satellite PRN code because it has very low cross-correlation properties and good autocorrelation, the sequences are generated by using two feedback shift registers and have specific pseudorandom numbers. The two 11-bit shift registers are used and called G1 and G2, the maximum length pseudo-noise (PN) codes are generated with a length of 2 ^11 = 2,046 bits. The NH code generator circuit is shown in Fig 2.1 the two-shift register polynomial and initial state for the NH code is as follow[31]:

Table 2. 5

BeiDou Code Generator and Initial States

Register	Polynomial	Initial State
NH code G1	$1+X+X^2+X^7+X^8+X^9+X^{10}+X^{11}$	01010101010
NH code G2	$1+X+X^2+X^3+X^4+X^5+X^8+X^9+X^{11}$	01010101010

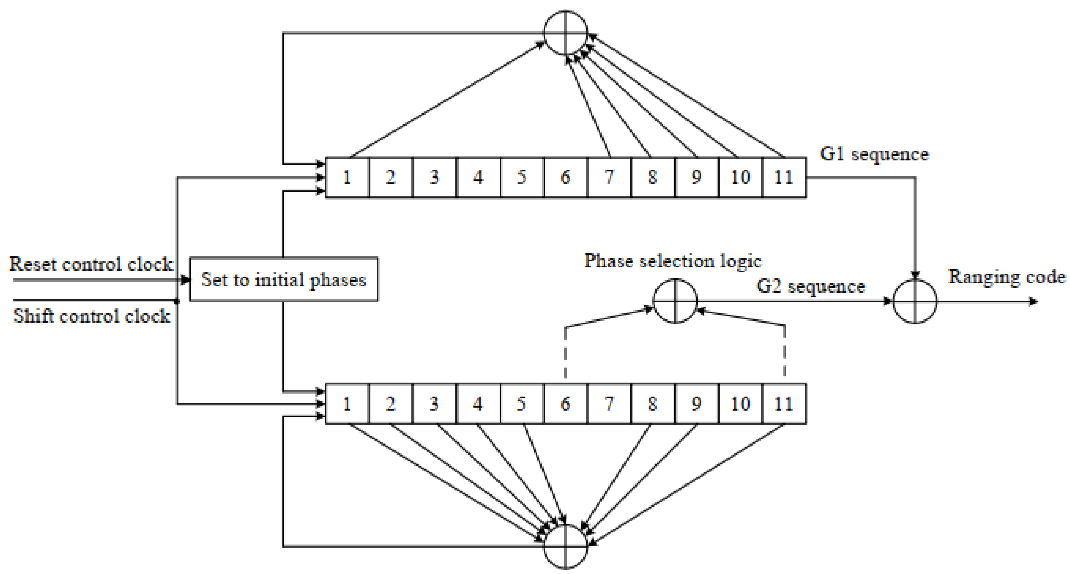

Fig 2. 3 BeiDou Ranging Code Generator

The different phase shift of the G2 sequence is accomplished by respective tapping in the shift register generating G2 sequence by the means of modulo-2 addition of G2 with different phase shift and G1 a ranging code is generated for each satellite. The phase assignment of the G2 sequence is shown in Table 2.6

Table 2. 6

Phase assignment of G2 sequence

SV ID	Ranging code number*	Satellite Type	Phase assignmentG2 sequence
1	1	GEO Satellite	$1\oplus3$
2	2	GEO Satellite	$1\oplus4$
3	3	GEO Satellite	$1\oplus5$

4	4	GEO Satellite	1⊕6
5	5	GEO Satellite	1⊕8
6	6	MEO/ IGSO satellite	1⊕9
7	7	MEO/ IGSO satellite	1⊕10
8	8	MEO/ IGSO satellite	1⊕11
9	9	MEO/ IGSO satellite	2⊕7
10	10	MEO/ IGSO satellite	3⊕4
11	11	MEO/ IGSO satellite	3⊕5
12	12	MEO/ IGSO satellite	3⊕6
13	13	MEO/ IGSO satellite	3⊕8
14	14	MEO/ IGSO satellite	3⊕9
15	15	MEO/ IGSO satellite	3⊕10
16	16	MEO/ IGSO satellite	3⊕11
17	17	MEO/ IGSO satellite	4⊕5
18	18	MEO/ IGSO satellite	4⊕6
19	19	MEO/ IGSO satellite	4⊕8
20	20	MEO/ IGSO satellite	4⊕9
21	21	MEO/ IGSO satellite	4⊕10
22	22	MEO/ IGSO satellite	4⊕11
23	23	MEO/ IGSO satellite	5⊕6
24	24	MEO/ IGSO satellite	5⊕8
25	25	MEO/ IGSO satellite	5⊕9
26	26	MEO/ IGSO satellite	5⊕10
27	27	MEO/ IGSO satellite	5⊕11
28	28	MEO/ IGSO satellite	6⊕8
29	29	MEO/ IGSO satellite	6⊕9
30	30	MEO/ IGSO satellite	6⊕10
31	31	MEO/ IGSO satellite	6⊕11
32	32	MEO/ IGSO satellite	8⊕9
33	33	MEO/ IGSO satellite	8⊕10
34	34	MEO/ IGSO satellite	8⊕11
35	35	MEO/ IGSO satellite	9⊕10
36	36	MEO/ IGSO satellite	9⊕11
37	37	MEO/ IGSO satellite	10⊕11
38	38	MEO/ IGSO satellite	1⊕2⊕7

39	39	MEO/ IGSO satellite	1⊕3⊕4
40	40	MEO/ IGSO satellite	1⊕3⊕6
41	41	MEO/ IGSO satellite	1⊕3⊕8
42	42	MEO/ IGSO satellite	1⊕3⊕10
43	43	MEO/ IGSO satellite	1⊕3⊕10
44	44	MEO/IGSO Satellite	1⊕4⊕5
45	45	MEO/ IGSO satellite	1⊕4⊕9
46	46	MEO/ IGSO satellite	1⊕5⊕6
47	47	MEO/ IGSO satellite	1⊕5⊕8
48	48	MEO/ IGSO satellite	1⊕5⊕10
49	49	MEO/ IGSO satellite	1⊕5⊕11
50	50	MEO/ IGSO satellite	1⊕6⊕9
51	51	MEO/ IGSO satellite	1⊕8⊕9
52	52	MEO/ IGSO satellite	1⊕9⊕10
53	53	MEO/ IGSO satellite	1⊕9⊕11
54	54	MEO/ IGSO satellite	2⊕3⊕7
55	55	MEO/ IGSO satellite	2⊕5⊕7
56	56	MEO/ IGSO satellite	2⊕7⊕9
57	57	MEO/ IGSO satellite	3⊕4⊕5
58	58	MEO/ IGSO satellite	3⊕4⊕9
59	59	GEO Satellite	3⊕5⊕6
60	60	GEO Satellite	3⊕5⊕8
61	61	GEO Satellite	3⊕5⊕30
62	62	GEO Satellite	3⊕5⊕11
63	63	GEO Satellite	3⊕6⊕9

2.7 Navigation Data

The Basic property of the BeiDou NH code is a navigation message. This message contains wide information that is used by the receiver to optimize the acquisition of satellite signals and calculate the user’s position. In this navigation, NAV message has data which is unique to the transmitting satellite and as well as common to all satellite. The data contain the information about the time of transmission of NAV message, a Hand over Word (HOW), clock correction, ephemeris and the health data of the transmitting satellite, almanac and the health data f all the satellites, coefficients for the ionospheric delay model and coefficient to calculate UTC.

BeiDou navigation message is formatted in D1 and D2 which is based on their rate and structure. Basic navigation information is in the D1 navigation message (broadcasting satellite fundamental NAV information, almanac information for all satellites as well as the time offset from other systems) and it is broadcast by MEO and IGSO satellites. The D2 NAV message contains basic NAV and augmentation services information (the BDS integrity, differential, and ionospheric grid information) and is broadcast by GEO satellites. The 50Hz navigation message is superimposed on both the Ranging code and the NH code. This navigation message is extracted by a 50bps BPSK demodulator which follows the NH code correlator. The narrow bandwidth of the navigation message ensures a high SNR ratio at the demodulator input and correspondingly low probability of bit error in the navigation error in the navigation message[33].

2.7.1 D1 NAV Message Frame Structure

The format D1 navigation message is structured in a super frame, frame, and sub-frame. A super frame has 3600 bits and lasts 12 minutes. A super frame consists of 24 frames (pages) and every frame is 1500 bits long and lasts 30 seconds. A-frame has 5 sub-frames and each subframe has 300 bits and lasts 6 seconds. Each sub-frame is consisting of 10 words and each word has 30 bits and lasts 0.6 seconds.

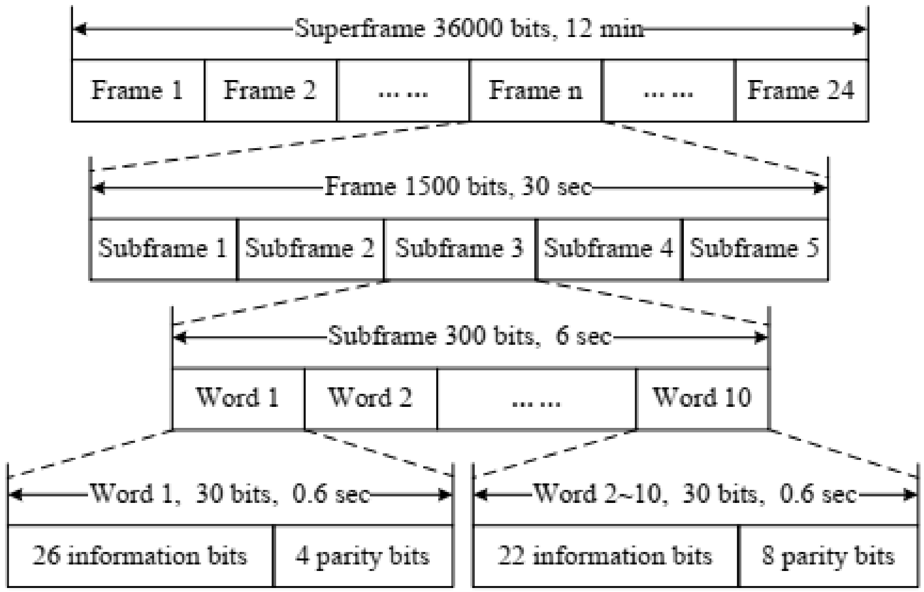

Fig 2. 4 Frame structure of NAV message in Format D1

Each word in the frame has contained NAV message data and the parity bits. In each subframe the first word, the first 15 bits are not encoded, and these 11 bits are encoded in BCH (15, 11, 1) for error correction.

2.8 Signal specifications

2.8.1 Satellite signal structure

The satellite signal is composed of the carrier frequency, ranging code, and navigation message. The ranging code and navigation message are modulated on carrier. To explain the signal structure the B1i is expressed as follow:

$$S_{B1i}^{j}(t) = A_{B1i} C_{B1i}^{j}(t) D_{B1i}^{j}(t) \cos(2\pi f_1 t + \varphi_{B1i}^{j}) \tag{2.1}$$

Where

Superscript j: satellite number.

A_{B1i}: amplitude of B1i.

$C_{B1i:}$ ranging code of B1i:

D_{B1i}: data modulated on ranging code

f_1: carrier frequency of B1i

$Ø_{B1i}$: carrier initial phase of B1i.

2.8.2 Noise and Interference

BeiDou RF signal is affected by noise and interference which affects the SNR ratio. The most basic noise is thermal noise which is denoted by N and is calculated by using formula $kTB_{n.}$ this will be discussed in detail in the simulation section.

The interference experienced by the BeiDou RF signal rather it will be intentional or unintentional can be summarized in Table 2.7.

Table 2. 7

Types of RF Interference and Sources

Sr No	Type	Source
1.	Wideband Gaussian	Intentional noise jammers
2.	Wideband phase / frequency modulation	Television transmitter's harmonics
3.	Wideband spread spectrum	Intentional spread spectrum jammers
4.	Wideband pulse	Radar transmitters
5.	Narrow band phase / frequency modulation	AM stations transmitter's harmonics

The source of error that effect the accuracy of the BeiDou receiver are satellite clock error, Ephemeris error, ionospheric delay, tropospheric delay, receiver noise, and Multipath effect.

2.9 SAR (Synthetic Aperture Radar)

Synthetic Aperture Radar is an echo mode array imaging system. In SAR, a long antenna is needed for this purpose a signal processing technique is used to generate an effective long antenna rather than the use of an actual long physical antenna. It provides a multidimensional database that can be manipulated with the aid of signal processing methods to obtain an image that carries information on the target area under study. SAR can be two types: monostatic, where the same antenna is used for transmission and reception purpose, where in the case of bi-static SAR separate antenna are used for transmission and reception[34]. The bi-static SAR principle is traditionally based on the radar positioned on an airborne platform while the image object is static or moving. When the radar moves, the reflected signal from the same imaged object is processed to synthesize an antenna with a synthetic aperture.

2.9.1 SAR Imaging

As stated above SAR is a two-dimensional image product. One dimension is cross-track and is called range, it measures “line of Sight” distance from the transmission of a pulse to receiving the reflection from a target. The other dimension is called azimuth (or long track) and is perpendicular to the range. To obtain fine azimuth resolution, a physical antenna is needed to focus the transmitted and received energy into a sharp beam. A large aperture is required to obtain fine imaging resolution (several hundred meters are often required). However, airborne radar could collect data while flying this distance and then process the data as if it came from a physically long antenna. The distance the aircraft flies in synthesizing the antenna is known as the synthetic aperture[35].

2.10 Doppler Frequency

The received signal frequency at the receiver is different from the transmitted signal frequency by a Doppler frequency offset and it is due to target or receiver motion relative SV. The receiver computed the satellite velocity utilizing ephemeris data and the orbital model that is present in the receiver. Satellite moving towards or away from the receiver, the received frequency is changed. Satellite at the closest point to the receiver the doppler frequency shift is zero[22].

2.11 Matched Filter

The peak signal power to average noise power in a matched filter maximizes its ratio of the peak. The filter must change if the signal is changed so that to match the signal being used. A matched filter is designed to provide a maximum signal to noise ratio at its output for a transmitted signal.

If $f(t) \leftrightarrow F(\omega)$ is the frequency response of the input signal then a network of frequency response $h(t) \leftrightarrow H(\omega)$ is matched to the input signal if $H(\omega) = F^*(\omega)$ or $\{h(t) = f(-t)\}$. In the presence of white noise, it will maximize the peak output signal voltage to RMS output noise voltage that is $2E / N_o$. Therefore, the maximum signal to noise ratio will depend on the input signal energy and the power spectral density of the noise rather than the shape of the waveform used. The impulse response of the matched filter is h(t) ↔ F*(ω) or h(t) = f(-t) i.e. h(t) is the time reverse of the matched input signal f(t). The output of matched filter g(t) will be: -

$$g(t) = f(t) * h(t) = f(t) * f(-t) = f(x) f(x-t) d \int x \qquad (2.2)$$

Hence the equation depicts the autocorrelation function of f(t).

The term matched filter is often used synonymously with the product integrator or correlator. However, it must be understood that the correlator output and the matched filter output are equal only at sampling instant (t=T) as shown below in Fig 2.5.

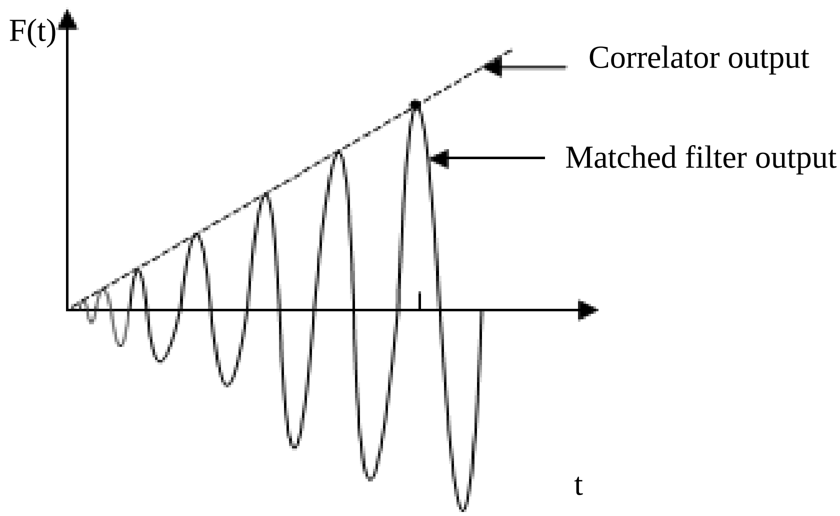

Fig 2. 5 comparison of matched filter and correlator

It can be summarized that the basic characteristic of the matched filter is that its impulse response is a delayed version of the mirror image (rotated on the T=0 axis) of the input signal waveform.

2.12 FM Chirp Signal

A long duration pulse is required for good detection in the radar environment to increase signal energy. A short duration pulse is required to improve target resolution (a term used to imply that two or more targets are separated to perform necessary measurement). With the help of FM, a wave form can be designed to have both short duration pulse and long duration pulse. This can be achieved in conjunction with a matched filter and the signal processing technique is called pulse compression. The linear FM pulse is called a chirp signal and can be represented by the following simplified equation for a linear increase of frequency with time.

$$f(t) = \cos(\omega_0 t + \pi \, {}^{F}\!/\!{}_{T} \, t^2) \; for \; -{}^{T}\!/\!{}_{2} < t < {}^{T}\!/\!{}_{2} \qquad (2.3)$$

where F= f_2 -f_1 is the sweep bandwidth with f_1 and f_2 being the initial and the final instantaneous frequencies and T is duration as depicted in Fig 2.6

Chirp signal (linear frequency modulation) pulsed signal is the preferred transmitted signal used to obtain a wide frequency band within a relatively wide time

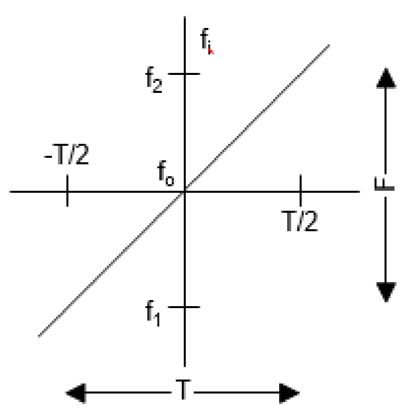

Fig 2. 6 Chirp Signal

width. By using chirp signals high peak power can be avoided and hence reducing the complexity of the radar system.

2.13 Target Characteristic

When a signal from a satellite strike with the target it reflects in various directions. The reflection nature depends on the nature and characteristics of the target.

The Radar cross-section for the reflecting objected is stated as the ratio of the power density of the signal reflected in the direction of the receiver to the power density of the radio wave incident upon the target. Thus, Radar Cross Section is to measure the target ability to reflect signals in the direction of the receiver. RCS unit is the square meter and measurement for the signal reflected from the target are made in comparison to the signal reflected from a perfectly smooth sphere having a cross-sectional area of one m^2.

The point target resolution theory is applied to facilitate the simulation processes. The target is modeled as a set of point scatters. A target can be resolved from another target if it falls in a different bin (or cell) and its return is not marked by self-clutter. The theory for the resolution of the points target states that the transmitter signal must have adequate bandwidth to achieve the desired small range resolution cell and it must be coherently integrated over a long enough time to achieve the small Doppler resolution cell. In the simulation it is assumed that the target act like point target dan does not experience power deterioration due to their RCS nor do they depend on the angle of incidence of the reflected signal[5].

3. BeiDou Receiver

It is preliminary to understand satellite receiver a brief overview of satellite signal period acquisition is presented below: -

3.1 Satellite signal acquisition

The BDS receiver has four satellites in view and the receiver must find the starting time of the unique NH code for each satellite in view. The receiver correlated the received signal with the replicated NH code generated in the receiver. The starting time of receive signal code is not known when the receiver initiates the locking process, a random starting point is selected. The received code is compared with the code generated in the receiver, bit by bit, through all 2046 bits of sequence, either lock is found, or the receiver does not have correct code for the satellite signal being received. All this process takes a minimum of 1 second to search all 2046 bit position of 2046 bit NH code, the receiver can take 15 seconds to acquire the first satellite in a normal case. After getting the first satellite code lock, then the other remaining satellite can acquire in few seconds because their IDs are known from the data transmitted in the signal of the initially acquired satellite.

Before the correlation proceeding receiver can find at least one satellite Doppler frequency offset. The bandwidth of the receiver can match the bandwidth of the NH code. The NH code noise bandwidth receiver is 2.046MHz and the velocity of the satellite is 2.926km/s.

As stated in the above section BeiDou receiver perform a search for satellite signal in two dimensions. Time is the first dimension i.e. the code and the frequency is the second dimension. After performing the search, the receiver establishes a tow tracking loop, for the frequency domain established phase lock loop and the time lock loop in the 2046-bit code space domain, with the goal of both tracking the code and carrier phases for the signal. It can start extracting data from the signal once it acquires a lock on the first satellite [36]. To perform a parallel search modern receiver, boast multiple physical channels. The code generator and the tracking oscillator are mostly implemented in VLSI DSP chips and the tracking loops are closed in software by the receiver's computer.

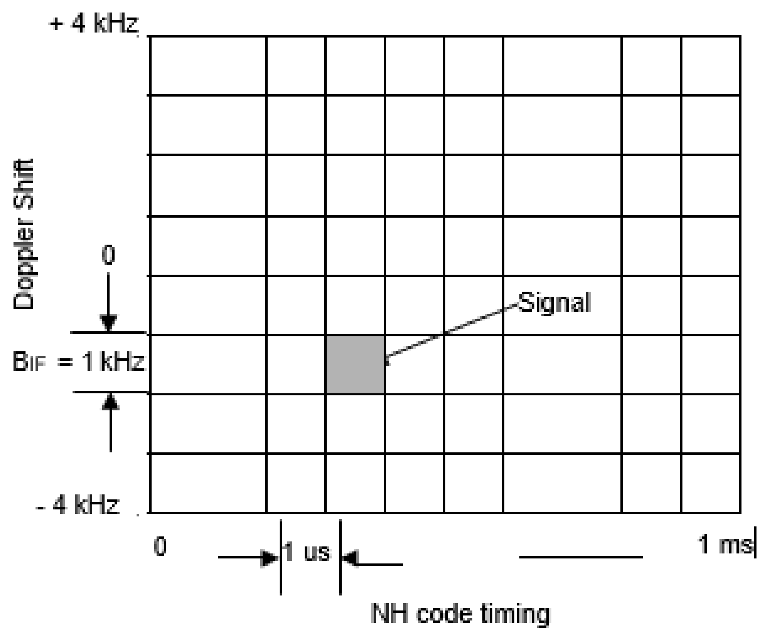

Fig 3. 1 Code Synchronization and Doppler Tracking Matrix

3.2 BeiDou Receiver operation

Details of the BeiDou Receiver are explained below.

3.2.1 Selection of satellite

The receiver checks which satellite is visible for tracking purposes, then the satellite tracking sequence begins. Receiver target a visible satellite to track and begin the acquisition process. Initially, first satellite tracking is relatively time consuming, and after that remaining satellite will be easily tracked. After tracking the first satellite successfully the Rx can demodulate the navigation message data stream and acquire the latest almanac as well as the health status for all satellites in the constellation[36].

3.2.2 Acquisition of satellite signal

The satellite signal acquisition is explained in section 3.1. The receiver code is measuring the delay in the transit time of the signals between the satellite and receiver and hence, the range between the satellite and receiver position.

3.2.3 Signal down conversion

The RHCP (Right Hand Circularly polarized) antenna with hemisphere gain pattern received RF (Radio Frequency) signals from all BeiDou SV in view. To improve the SNR ratio of the signal, the signals are amplified with a low noise pre-amplifier. This amplified RF conditioned signal is then down converted to an intermediate frequency (IF) using signal mixing frequency from a local oscillator (LO). For down conversion, sometimes more than one down conversion stages are employed to convert two intermediate frequency (IF) down to a frequency near the code baseband that is sampled by an analog to digital converter. In-phase (I) and quadrature-phase (Q) digital samples are taken to secure the phase information in the received signal. This process is shown in the Fig 3.2.

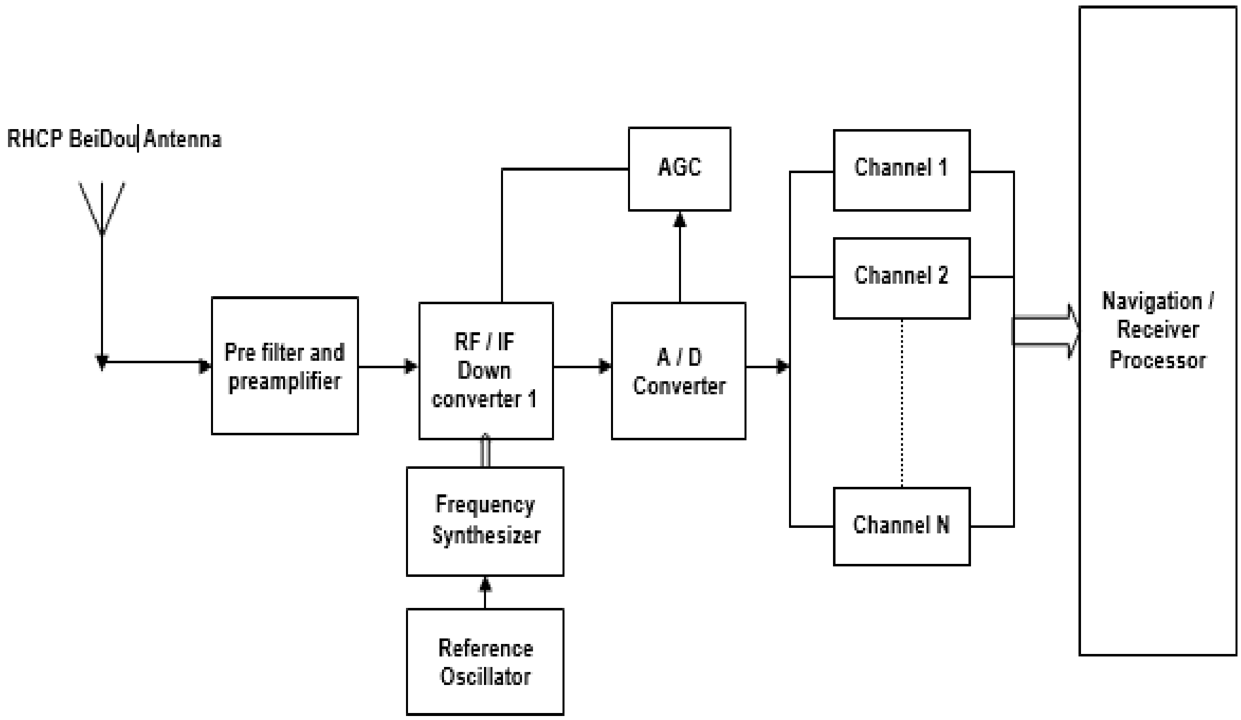

Fig 3. 2 Generic Digital BeiDou Receiver Block Diagram

3.2.4 Signal Code Tracking

The pseudo-range measurement between the BeiDou satellite and the BeiDou receiver code tracking loop is used. In the receiver, the replica of the code of the targeted satellite is generated with the receiver's code tracking loop. Initially the code synchronization is maintained by correlating the received signal with half chip early and late codes. For the achievement of a precise frequency lock to the incoming signal, the carrier is reconstructed, by using doppler measurement from the carrier tracking loop, the center frequency of the replica code is adjusted. Thereby allowing more precise pseudo-range measurements[16].

3.2.5 Signal carrier tracking and detection of data

At the reception, satellite carrier tracks by adjusting the frequency synthesizers, a stationary phase is produced at the output of the code tracking loop. The carrier phase and Doppler can be calculated by using the in-phase and quadrature-phase component. In the detected signal a data bit is detected by the sudden change in phase. The receiver estimates the velocity between the satellite and the receiver from the doppler change. For the receiver velocity in three-dimension, receiver can use the doppler measurement from four (or more) satellites.

3.2.6 Demodulation of data

Navigation message once locked by the carrier tracking loop; data can be extracted. The receiver can enable the detection of the starting of each sub-frame of the navigation message begins with a preamble, contained in the telemetry word.

Fig 3.2 shows the receiver complete block diagram and each channel in the receiver, and further elaborated with the aid of Fig.3.3. The digitized signal is applied to the input, carrier tracking loop, and code associated function are illustrated for simplicity. The digital IF is strip off the carrier (plus carrier Doppler) as a first stage with the replica carrier (plus carrier Doppler) signals to produce in-phase(I) and quadrature-phase (Q) sampled data. The output of the in-phase mixers I signal would be mostly multiplied with thermal noise by the replica digital sine wave (to match the SV carrier at IF). Similarly, the output of the quadrature-phase mixers Q signal would be mostly multiplied with thermal noise by the replica digital cosine wave (to match the SV carrier at IF). The SV desired signal buried in noise until the I and Q signal is collapsed to baseband by the code stripping process.

In phase, I and quadrature-phase component Q had produced, and these are 90^0 phase apart, the amplitude of the resultant signal can be computed from the vector sum of I and Q components, and the phase angle can be determined from the arctan of Q/I. codes can be observed to be rotating in skew motion when observed. The I and Q signal is then correlated with the early, late, and prompt replica code (plus code Doppler) synthesized by the code generator as shown in Fig 3.3. A 3-bit shift register and a code NCO. Replica code phase misalignment with respect to incoming SV code produces a difference in the magnitude vector of the early and late correlated outputs. So that from code tracking loop the magnitude and direction of the phase change can be detected and corrected[37].

GNSS receiver can be broadly classified into two categories. A receiver that continuously tracked, that has five or more hardware channels to simultaneously track four satellites plus other channels to acquire new satellites. A sequential receiver tracks the necessary satellite by typically using one or two hardware channels.

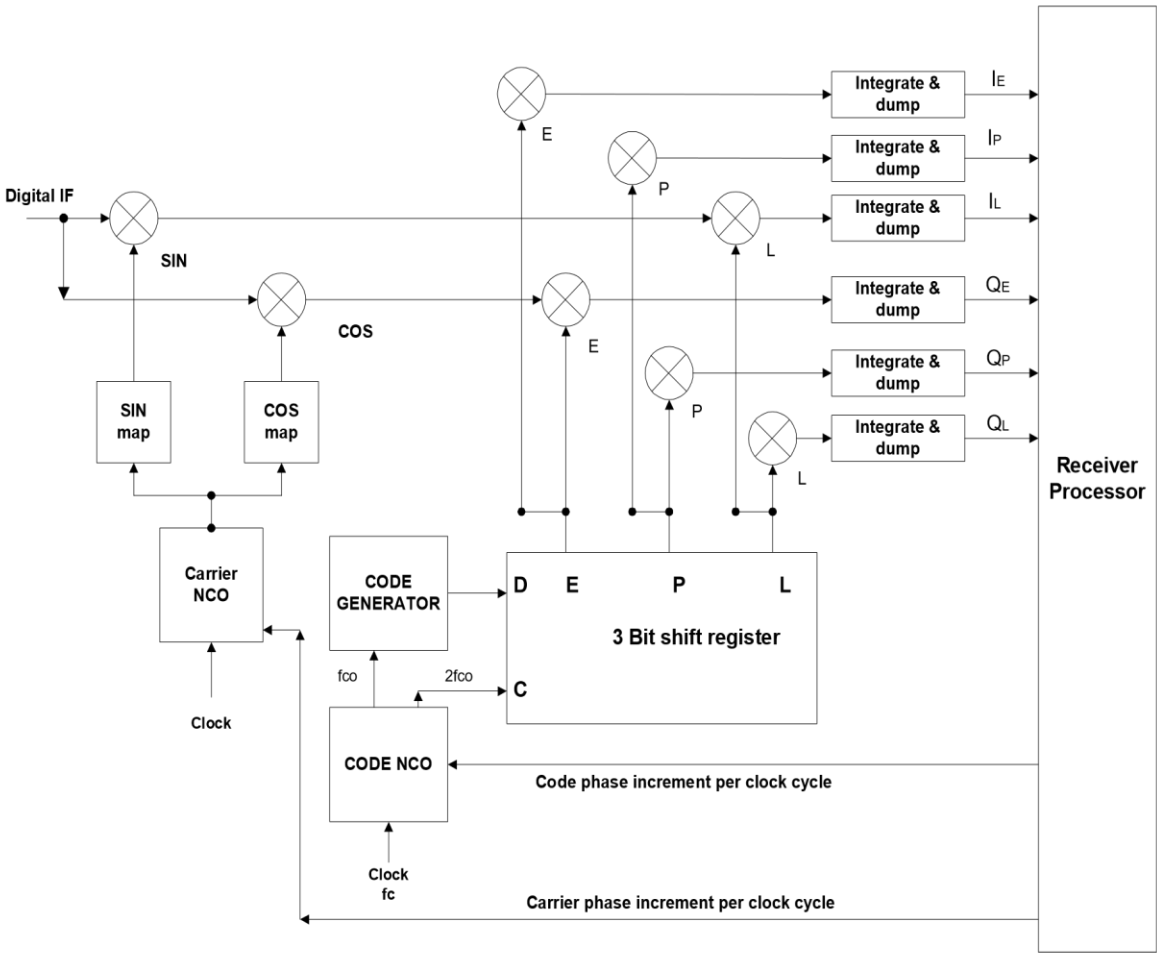

Fig 3. 3 BeiDou Receiver Channel Block Diagram

Calculation error is due to noise and it can corrupt the measurement process in receiver. During the measurement process errors induced due to noise in the ionospheric correction and system dynamics not considered. (e.g., user clock drift). A Kalman filter characterizes the noise source to minimize their effect on the desired receiver outputs.

3.3 Modified BeiDou receiver

A conventional GNSS receiver cannot be used for practical implementation of the system. The proposed and modified BeiDou receiver’s high-level functional block diagram is shown in Fig 3.4

A RHCP antenna receives the direct signal from the satellite and down-converts as stated in section 3.2.3 above. The delay in code and code offset are calculated and performed a similar way that of a conventional BeiDou receiver. With suitable electronics, the direct signal is converted to baseband, and I and Q component are extracted. It is essential to reduce the power of direct signal to an appropriate factor so that it becomes comparable to the indirect signal in terms of amplitude. This process is deemed necessary to make sure that the indirect signal is not lost in the tails of the autocorrelation function of the direct signal. For the direct signal attenuation, some suitable method or some cancellation technique is required for the main beam. Similarly, the LHCP antenna receives the indirect signal and then down-converts to IF and later sampled to convert in digital format by conventional means. The reflected received signal is unfortunately very weak and buried in noise, at the receiving end a high-efficiency low noise preamplifier will ensure that the SNR ratio is acceptable enough to effectively perform the code delay and code offset calculation. Circuit for code offset calculation must be designed so that both direct and indirect signals must be synchronized with each other for the necessary measurement. With suitable electronics, the direct signal is converted to baseband, and the in-phase(I) and quadrature-phase (Q) signal components are extracted. Match filtering technique compares the direct and indirect signals and with the appropriate means, the image is displayed. For later processing and signal analysis data can be stored in memory [29].

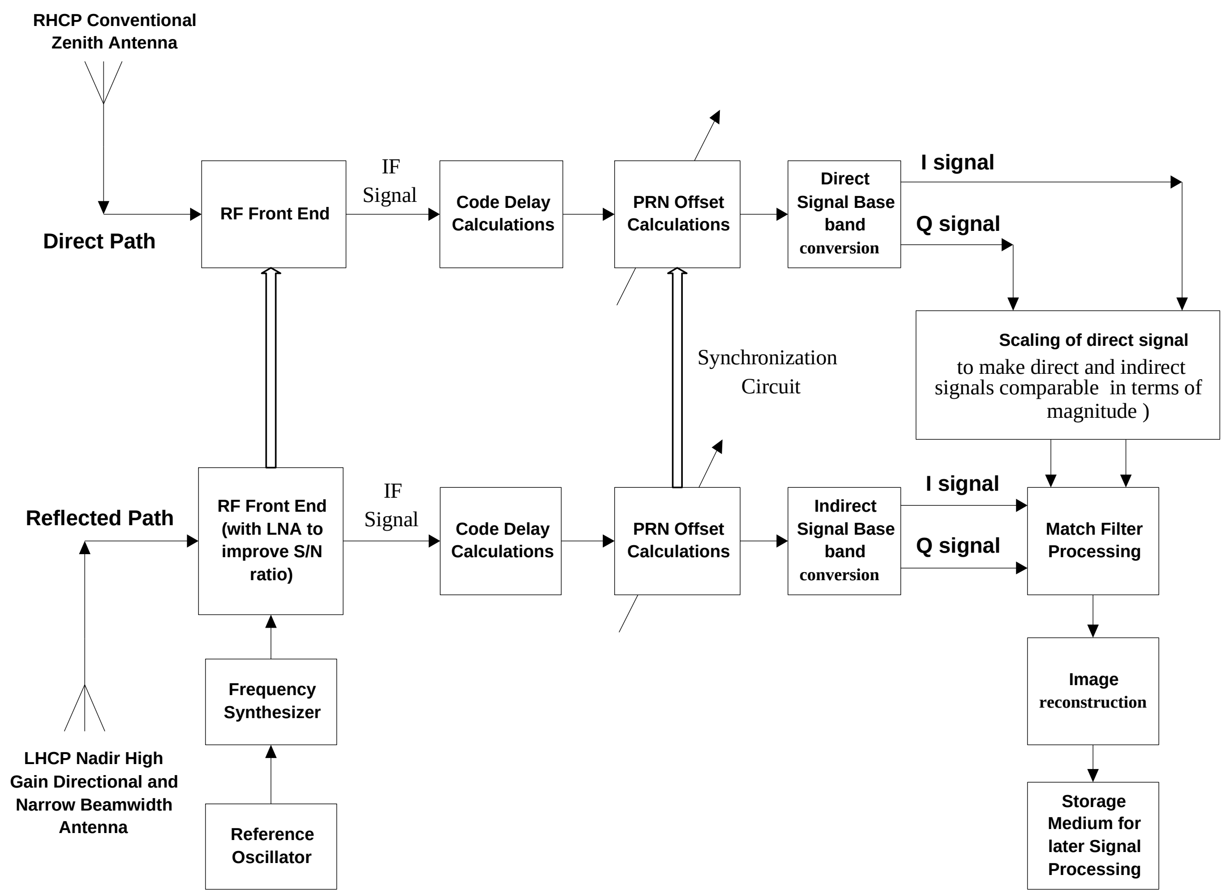

Fig 3. 4 Modified BeiDou Receiver

In software form implementation of a/m receiver part will be preferred. BeiDou signal structure and processing in-depth understanding is necessary to develop software BeiDou receiver. To analyze signals from the LHCP antenna (used for reflected signal reception), it will be practically more suitable. The modified software Rx can perform signal acquisition and tracking, navigation data extraction, code offset calculation, and direct signal scaling, and more importantly the parameters can be conveniently changed, and the receiver can be adapted to suit different conditions.

4. Analysis of Power Budget

In this research the direct BeiDou B1i signal and the reflected signal reception can be simulated to generate an image of the area of interest. For all these, the concept of SAR (Synthetic Aperture Radar) is used, and before all these things can be performed. A Power Budget Analysis is performed for BeiDou B1i Signal and GPS L1 signal.

In this analysis their suitability for remote sensing application is verified, also the accuracy and compatibility. For the research, the power Budget analysis is necessary and can be performed before the start of the research. The direct BeiDou signal strike with any object and reflect. This reflected signal can be acquired with and appropriate equipment’s. This hardware can equip on an arial object, to get the synthetic aperture effect, and required change in geometry. All this scenario of the acquisition of reflected signal depicted in the a/m Fig 1.1

The image reconstruction from BeiDou signal based on Bi-Static radar configuration, range resolution is an important parameter. The range resolution is given by.

$$\delta_r = \frac{c}{2B\cos(\frac{\beta}{2})} \qquad (4.1)$$

Where B is the Bandwidth of transmitted signal and β is bi-static angle. Range resolution is depending on bandwidth of transmitted signal and bi-static angle. For the better resolution of the image either improve the bandwidth of the signal or bi-static angle. In this research we have no control on the transmitter, then modified the bi-static angle. The maximum resolution can be obtained at zero angle at which bi-static radar become mono-static[38][39].

The GPS L1 civil signal is transmitted at center frequency of 1572.42 MHz with the coarse acquisition (C/A) code. The BPSK modulation C/A code has a chip rate of 1.023 with period of 1 millisecond. The bandwidth of L1 frequency signal is 2.046 MHz . Because of low bandwidth and inferior GPS L1 signal power detection of reflected signal is almost impossible using conventional signal acquisition methods. The BeiDou B1i signal is design for civilian purpose and freely available. Signal transmitted frequency is 1561.098MHz with chip-rate of 2.046MHz, period of 1milli second and 4.092 bandwidth. The BeiDou B1i signal Ranging length is 2-times longer as compared to GPS L1 signal and had better signal to noise ratio due to more bandwidth and spatial resolution then the GPS L1 frequency signal.

Improvement in signal structure will in turn improve position accuracy and availability of services. Power budget analysis will be performed at the initial stage of this research, and the simulation will be performed for the GPS L1 and BeiDou B1i signal and result are compared.

4.1 For GPS L1 signal power budget analysis

Initially GPS L1 signal analysis is performed for bi-static radar power budget, and then followed by BeiDou B1i signal. The signal transmitted power is represented with P_t and the gain of transmitter is represent with G_t , R_1 represent the range from transmitter to target and it is 2200Km. R_2 is represented the range from target to receiver and it called the slant range and is given by[40].

$$R_2 = \sqrt{\frac{Pt\ GtGr\lambda^2\sigma}{(4\pi)^3 R1^2 P_r}}\ km \qquad (4.2)$$

The power received at antenna of the receiver is given by[40].

$$P_r = \frac{Pt\ Gt}{4\pi R1^2 4\pi R2^2} * \sigma A_{ef} \qquad (4.3)$$

Where $A_{e.f}$ is receiver antenna effective area[40].

$$A_{ef} = \frac{\lambda^2 * Gr}{4л} \qquad (4.4)$$

The received power of reflected signal is also depends on the gain of the receiving and the cross-section of the radar. The greater RCS indicates that target can easily be detected, and the other on which detection of target also depends is target size, materials, and shape.

The signal to noise ratio for Bi-static radar and receiver is given by[40].

$$SNR = \frac{Pt\ GtGr\lambda^2\sigma}{(4\pi)^3 R1^2 R2^2 KTB_n} \qquad (4.5)$$

Where G_r is antenna gain of receiver, P_r is received power at receiver, the noise of the receiver is $N_r = kTB_n$ and the λ is GPS L1 signal wavelength. Where $R_1 = R_2$ the SNR is maximum however GPS satellite is transmitting signal at height of 2200Km. utilizing GPS as the transmitter of opportunity SAR system is suitable for the application of limited ranges. The target to receiver range is limited to 1000m. In the graph GPS L1 frequency SNR verses Range plot with 10 m^2 cross-section target is shown in Fig. From Fig SNR is very poor even for a short range. That is the basic limitation of this type of passive microwave imaging radar. In the area of interest, the target image is generated by correlating the signal for longer period from which get processing gain and improve the

SNR and thus makes it possible. The correlation can be performed in coherent and non-coherent summation manner, as noise is uncorrelated therefore the SNR is improved.

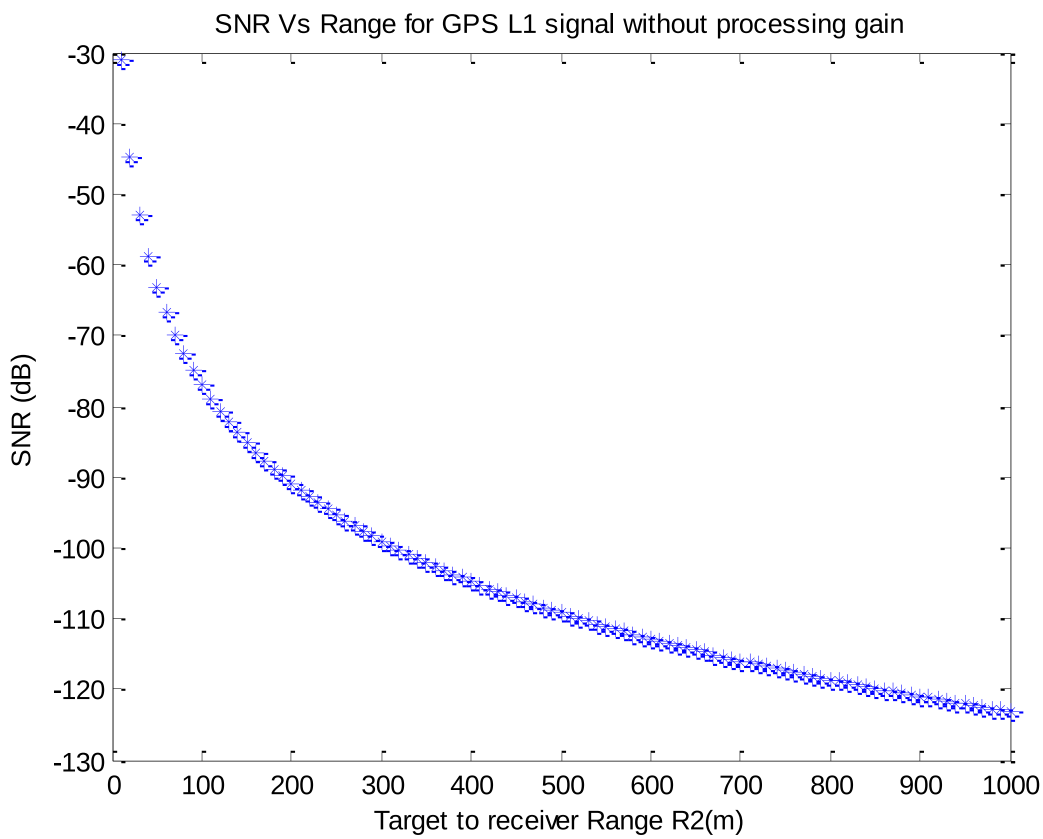

Fig 4. 1 SNR Vs Range for GPS L1 signal without processing gain

From Fig 4.1 shows GPS L1 signal simulation result for SNR verses range. Simulation was carried out from 0-1000m range and SNR for different ranges was calculated. The SNR is of -70dB on the range of 100m for GPS signal. That is very low SNR and detection is almost impossible. Correlating the signal for longer period the SNR is obtained is given by[41].

$$SNR = \frac{Pt\ GtGr\lambda^2\sigma}{(4\pi)^3 R1^2 R2^2 KTB_n} G_{sp} N^{0.8} \quad (4.6)$$

Where G_{sp} is processing gain and N is maximum number of non-coherent samples. Non-coherent summation is used as it takes larger values of N to obtain required gain then the coherent summation. In non-coherent summation longer period is used to recover the signal from noise and increases processed SNR.

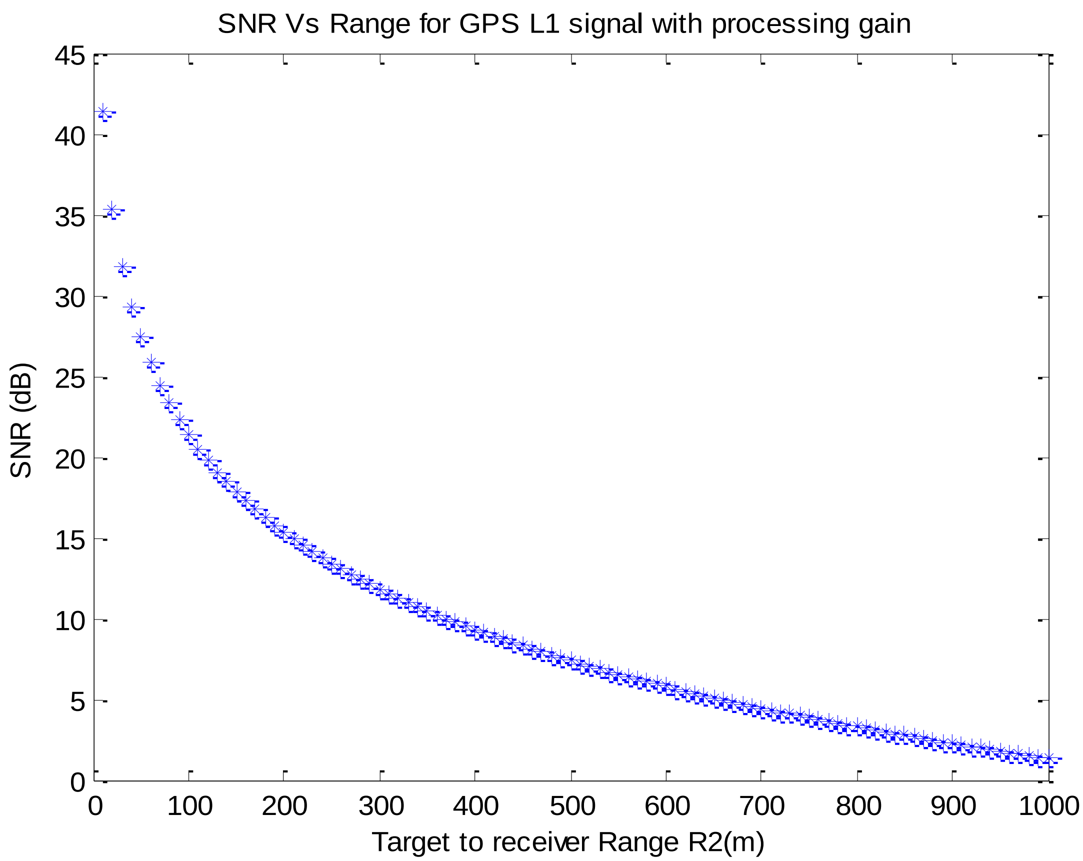

Fig 4. 2 SNR Vs Range for GPS L1 Signal with processing gain

The SNR vs range simulation result's for GPS L1 signal with processing gain of 43dB is shown in Fig 4.2 Correlate the signal for longer period with non-coherent summation has improved SNR and at range of 100m, the SNR is 22dB and therefore detection is possible and further utilization of reflected signal for remote sensing application can be carried out the parameter's values used for "the power budget analysis of GPS L1 signal" is summarized in Table 4.1.

Table 4. 1

Parameter values for analysis

No.	Parameter	Value	Unit
1	Transmitted power (Pt)	40	Watt
2	Transmitter antenna gain (Gt)	13	dB
3	Receiver antenna gain (Gr)	30	dB
4	Bandwidth	2.046	MHz
5	Receiver noise level (Nr)	-131	dB
6	Processing interval (TQ)	0.1	Second

4.2 BeiDou B1i signal power Budget analysis

BeiDou B1i signal is also used for remote sensing application and compare its results with GPS L1 signal. The power analysis for BeiDou B1i reflected signal in same scenario is performed and parameters used for power budget analysis of BeiDou B1i signals are summarized in Table 4.2.

Table 4. 2

Parameters Values for analysis

No.	Parameter	Value	Unit
1	Transmitted power (Pt)	43	Watt
2	Transmitter antenna gain (Gt)	15	dB
3	Receiver antenna gain (Gr)	30	dB
4	Bandwidth	4.092	MHz

5	Receiver noise level (Nr)	-131	dB
6	Processing interval (TQ)	0.1	Second

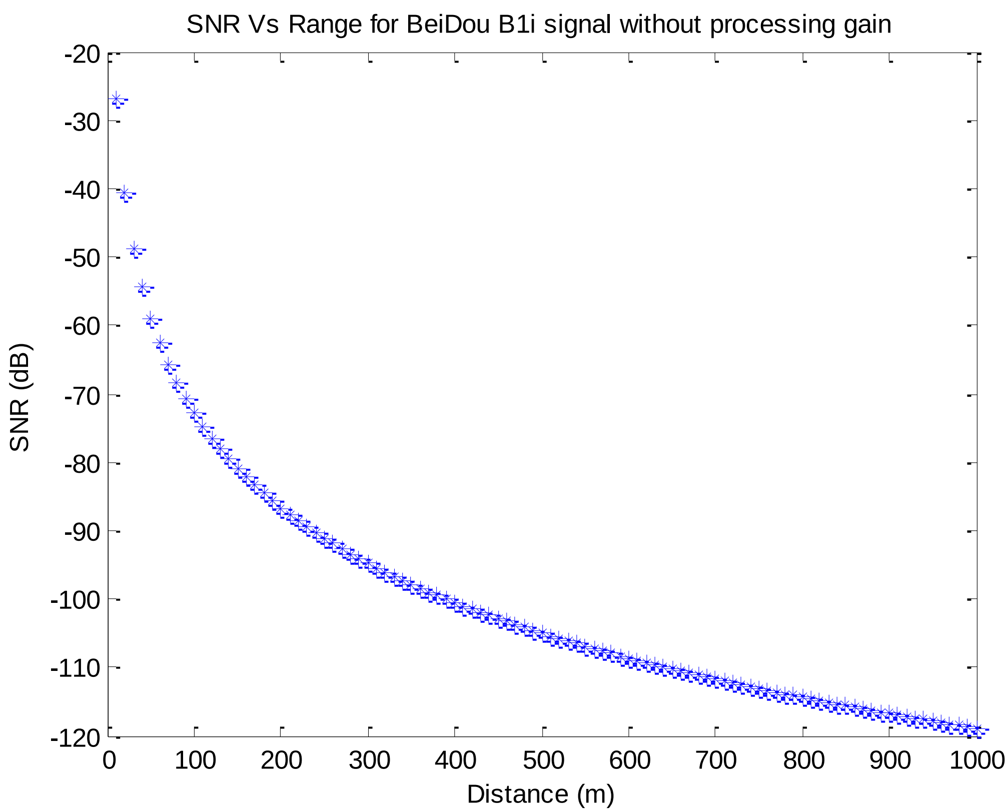

Fig 4. 3 SNR Vs Range for BeiDou B1i Signal without processing gain

The SNR vs range plot for BDS B1i signal for target cross-section of 10m is shown in Fig 4.3 While other parameters are same. For 0-1000 target range simulation is carried out and plot the SNR for these ranges. The SNR of received reflected signal from 100m is 25db which is higher from the GPS L1 signal.

The SNR can be more improved with processing gain by correlating for longer duration and is shown in Fig 5. The processing gain is given by Gsp=BTQ where B is bandwidth and TQ is processing interval.

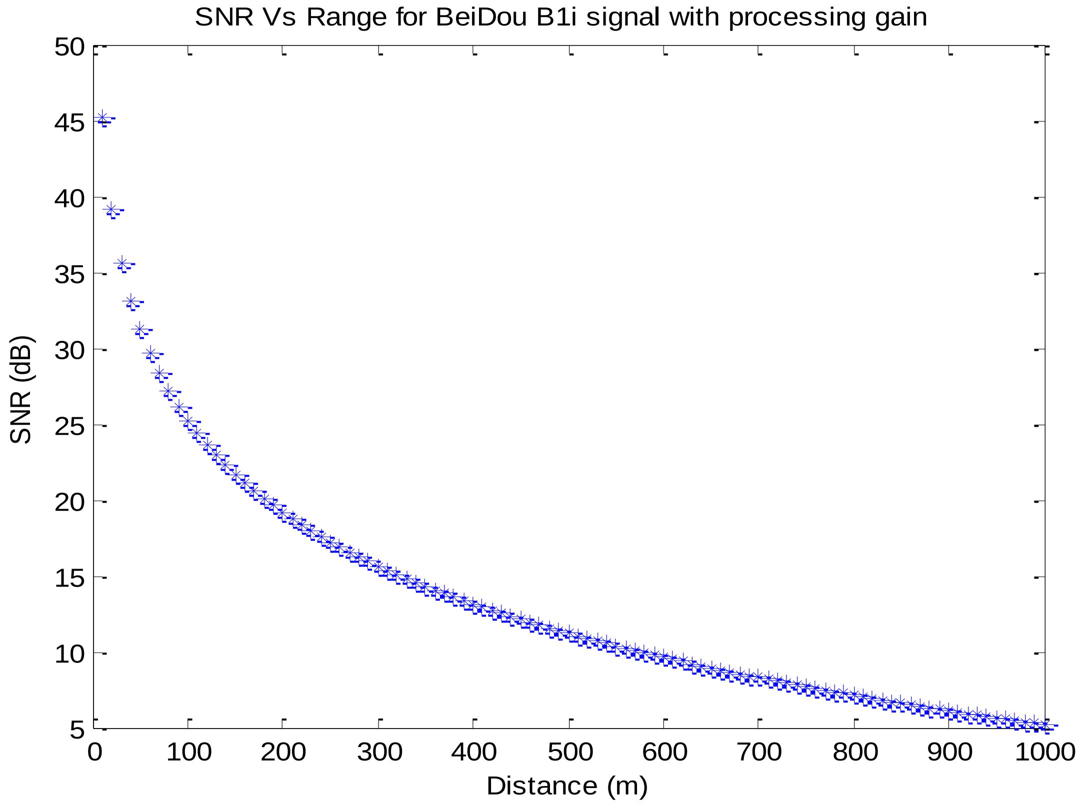

Fig 4. 4 SNR Vs Range for BeiDou B1i Signal with processing gain

Table 4. 3

SNR of GPS L1 & BDS B1i reflected signal Target to receiver ranges.

Sr.No	**Target to receiver range R2(m)**	**Without processing gain**		**With processing gain**	
		L1 SNR (dB)	**B1i SNR (dB)**	**L1 SNR (dB)**	**B1i SNR (dB)**
1	10	-31	-27	41	45
2	50	-63	-59	27	31
3	100	-77	-73	21	25
4	200	-91	-87	15	15
5	500	-109	-105	7	11
6	800	-118	-114	3	7
7	1000	-123	-119	1	5

For the B1i signal same target ranges is calculated and plotted their SNR in Fig 4.4 For 100m target range SNR of B1i signal with processing gain is 25dB which was 21 in case of GPS L1 signal. SNR improved by correlating reflected signal for longer time more then 4dB as compared to SNR of L1 signal.

Table 4.3 Summarized the reflected GPS L1 and B1i frequency signal SNR for different target to receiver ranges. Which is evident that the SNR is very poor even at short distances, that is the main restriction for this type of microwave imaging radar. The processing gain obtain by correlating the signal for longer period of time during reconstruction significantly improves the SNR and thus makes its possible to utilized the reflected signal for remote sensing applications or generate an image for the area of concern using suitable hardware and reconstruction algorithm.

5. Simulation

Software used for simulation work is MATLAB version 2014a. A BDS and (SAR) Synthetic Aperture Radar Based imaging system is demonstrated. The whole simulation model / scenario is depicted in Fig 4.1

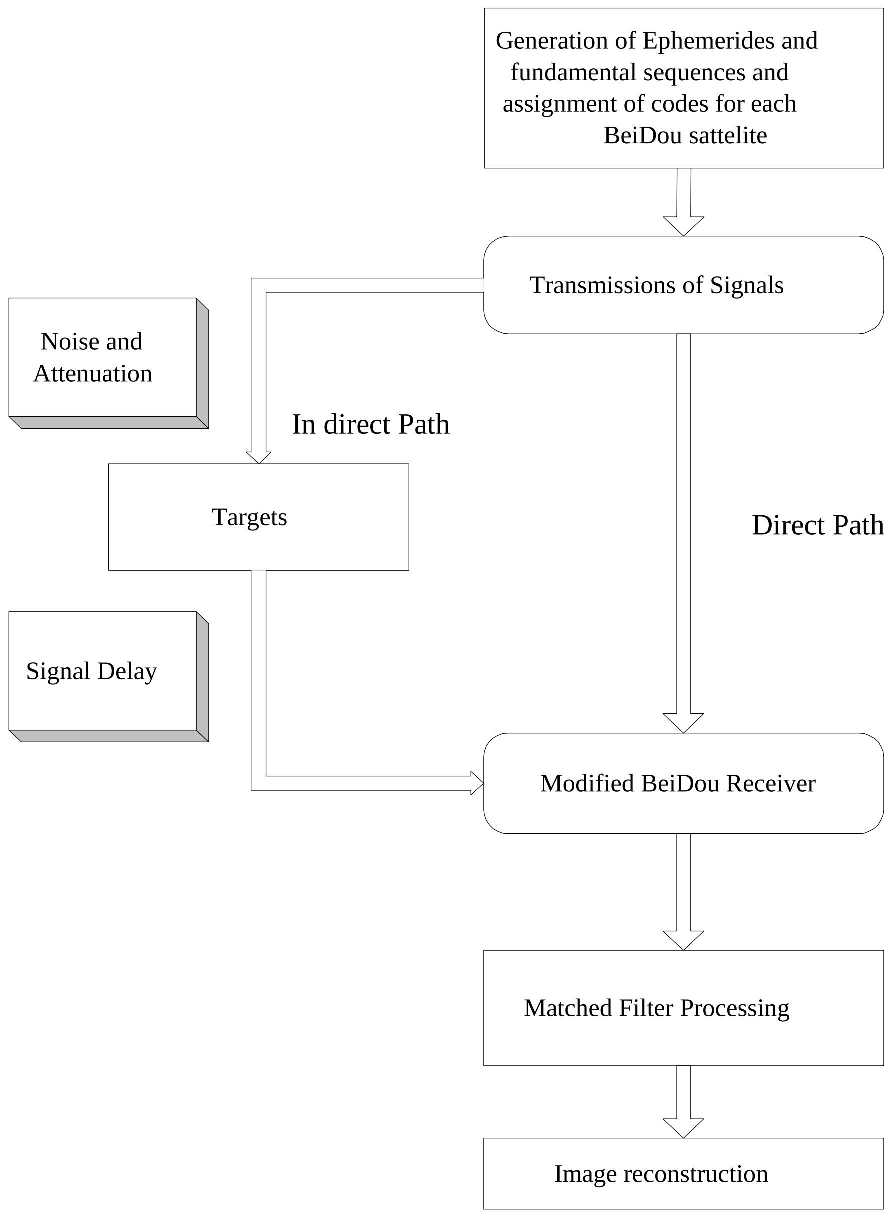

Fig 5.1 Block Diagram of Simulation

The idea of the bi-static radar principle and Bi-static radar technique is simulated for signal processing. Since the SV and Rx are moving, thus the signal obtained at the receiver position is a linear frequency modulated (FM) BeiDou or Chirp signal. The reflected signal received at the receiver after reflection from the target will also be approximately linear frequency modulated BeiDou signal. From the Radar concept, we know that range resolution is an inverse ratio to the signal bandwidth. Since BeiDou signal is a linear frequency modulated and thus has good property of pulse compression, which makes a multi-static SAR imaging system to obtain enough and good range and resolution, simultaneously. To further improve the system resolution a matched filter method is employed in the received signal processing.

From Fig 4.1 it's evident the generation and assignment of individual ranging code for each SV is simulated at the start of the program when these codes are received, the characteristics of these codes, effects along the way i.e. the delay and offset introduced are also simulated as close as possible. A matched filter concept is used, and a grayscale image is constructed. The signal is up-converted to RF frequency at the transmission end and the receiver end down-converted again to IF frequency. The up and down conversion is not simulated because it is cancelled during the modulation and demodulation process.

The simulation works have two broad parts. The simulation of the real-world i.e. the process that occurs in BeiDou satellite and transmission during space, these are the coordinate system, satellite orbits, code generation, and simulation of noise and attenuation etc. On the contrary simulation of the Beidou receiver entails the delay and code offset calculation, matched filter processing, and image reconstruction process.

5.1 Detailed MATLAB code description.

The MATLAB version 14a code and its constituent functions are given as Fig 5.2, the code is also elaborated in words.

We start the program by specifying Global variables, noise, sampling time, pixel size, and search area. The search area is set up to 2000x2000 meters in which the target can be located. The middle position of the search area is to be fixed on (0, 0, 6378000). A multi-dimensional array for the candidate position had three dimensions, one each for x, y, and z and z coordinates. Tow targets position are specified at (200, 0, 6378000) and (-200, 0, 6378000) respectively in BDSCS coordinate frame and the pixel size is set 20meters in the search area. The starting position of the receiver is specified to be at (300, 0, 0) m/s. As the SAR basic concept, the receiver is kept airborne with an altitude of 5 Km /hr. Nine satellite satellites have been selected; these are SV No. 7, 9, 14, 16, 22, 25, 26, and 27. The SV No.6 is allocated as the reference SV.

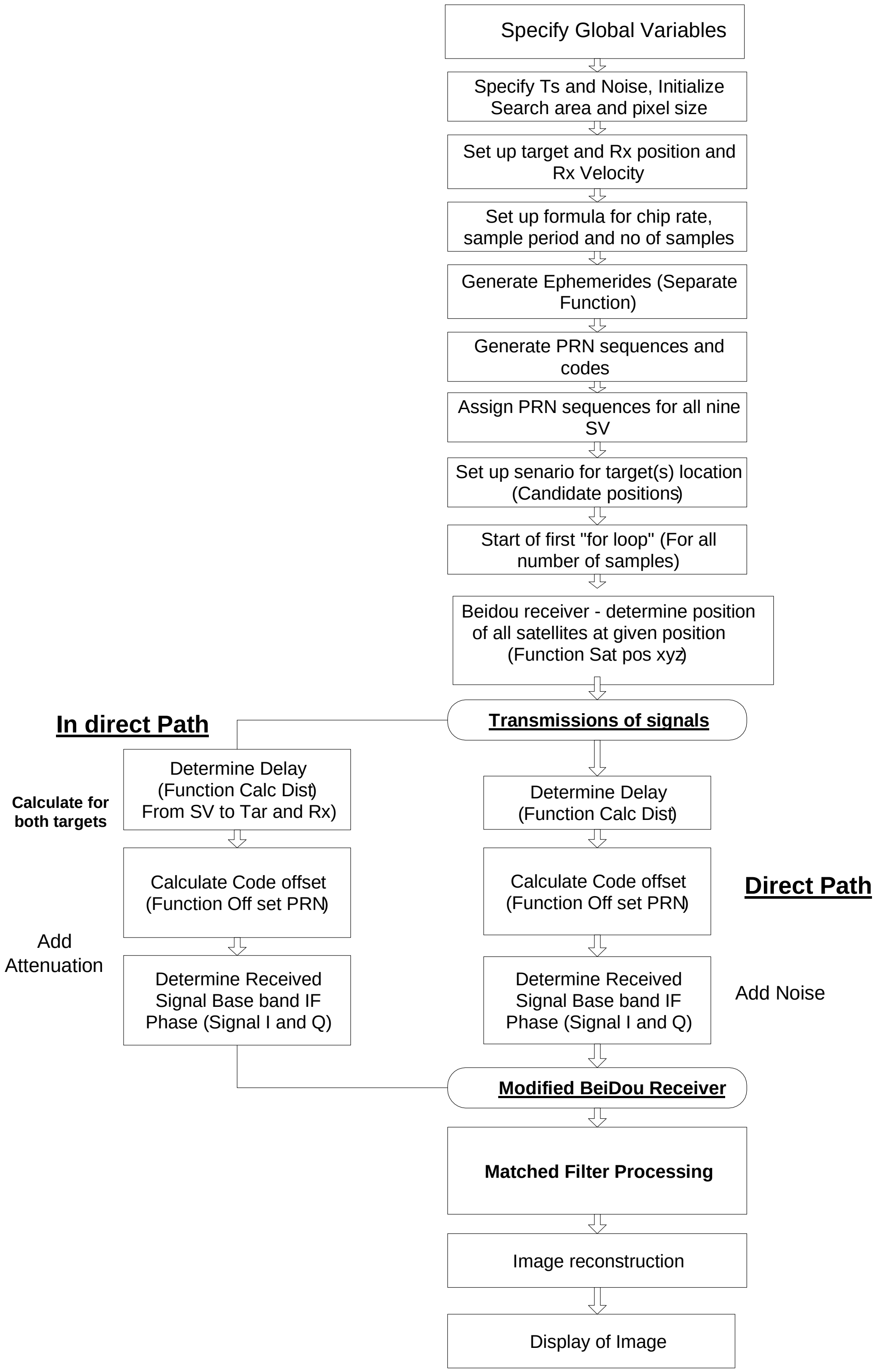

Fig 5. 2 Detailed Code Flow Chart

Table 5. 1

Computation of satellite position vector

Sr. No	Equation	Description
1	$\mu = 3.986004418 \times 10^{14} m^3 / s^2$	Geocentric gravitational constant of BDSCS
2	$\dot{\Omega} = 7.2921150 \times 10^{-5} rad / s$	Earth’s rotation rate of BDSCS
3	$\pi = 3.1415926535898$	Ratio of a circle’s circumference to its diameter
4	$A \square (\sqrt{A}\ )^2$	Semi major axis
5	$t_k = t - t_{0a}$ *	Time from ephemeris reference time
6	$n_0 = \sqrt{\frac{\mu}{A^3}}$	Corrected mean motion
7	$M_k = M_0 + n_0 t_k$	Mean anomaly
8	$M_k = E_k - e\mathrm{Sin}E_k$	Kepler’s equation for Eccentric anomaly (may be solved by iteration)
9	$\sin v_k = \frac{\sqrt{1-e^2}\sin E_k}{1-e\cos E_k}$ $\cos v_k = \frac{\cos E_k - e}{1-e\cos E_k}$	True anomaly
10	$\phi_k = v_k + \omega$	Argument of latitude
11	$r_k = A(1-e\cos E_k)$	Radius
12	$x_k = r_k \cos \phi_k$ $y_k = r_k \sin \phi_k$	Position in orbital plane
13	$\Omega_k = \Omega_0 + (\dot{\Omega} - \dot{\Omega}_e)t_k - \dot{\Omega}_e t_{0a}$	Corrected longitude of ascending node
14	$i = i_0 + \delta_i$ **	Inclination of reference time

15	$X_k = x_k \cos\Omega_k - y_k \cos i \sin\Omega_k$ $Y_k = x_k \cos\Omega_k - y_k \cos i \cos\Omega_k$ $Z_k = y_k \sin i$	Coordinate of the satellite antenna phase center in BDSCS

It is supposed that all signal power is reflected from the target towards the receiver. To reduce the receiver sensitivity oversampling is employed and A/D quantization noise thereby reducing the number bits required in A/D converter. For this simulation, the code has been oversampled 2 times to the chip rate of 2,046,000.

5.2 Calculation of satellite Coordinates

The code is to simulate the satellite ephemerides data (that contain the time of epoch, Keplerian orbital element at the time of epoch, and correction parameter) to calculate the velocity of SV and generate the position of the satellite in BDSCS coordinates

Table 4.1 depicts the algorithm with which the position vector of the satellite is computed. A program is written based on the table which calculates the position of satellites based on the input of time of week transit time and ephemeris data. The transit time is 70 msec and is approximately the time required for the signal to reach from satellite to the earth. The program is furnishing a vector of size three containing the (Xs Ys Zs) coordinate of SV in the BDSCS system.

5.3 Ranging code sequence Generation

Fundamental ranging code sequences are generated and assigned to the individual SV with separate ranging code for each satellite. The generation of code is undertaken in both types of simulation first by the SV and second by the receiver. In the receiver, the code has reproduced to perform the correlation process. The generation of code is already discussed in the receiver section. The code basic properties are having been elaborated in section 2.6.1. As stated in this section before mentioned there is two 11bit shift register with the help of which PRN sequence is generated, the code length is 2^{11}-2= 2,046 bits. The G1 first shift register feedback taps have connected to stage 1, 7, 8, 9, 10, 11, and feedback to stage 1. The G2i sequence is formed by delaying the G2 sequence by an integer number of chips. Each PRN sequence number is formed by selecting two tabs position on G2. Table 4.2 describes all these tabs combination which defines BeiDou PRN number and specifies the equivalent code shift chips.

MATLAB programming is used to generate the PRN code and different PRN code is assigned to each of nine SV. The codes are bi-phase which consisting a sequence of -1 and +1. These codes are repetitive in practical but for simulation purposes, it generated only once.

5.4 Signal delay calculation

The distance covered by the signal is calculated twice as from Fig1.1 in Fig one path is a direct path from SV to the receiver and the other is an indirect path from SV to target and then to the receiver. How to calculate the position of the satellite is explained in the above section. Elementary trigonometry is used to calculate the distance b/w satellite vehicle and receiver. The geometry of the satellite target and receiver is given in Fig 1.1

The D_{sr} is the distance b/w satellite vehicle and receiver and calculates with a simple formula.

$$D_{direct} = D_{sr} = D\sqrt{(X_s - X_r)^2 + (Y_s - Y_r)^2 + (Z_s - Z_r)^2} \qquad (5.1)$$

In A/M equation (Xs, Ys, Zs) represent the space vehicle coordinates that are in view and the receiver coordinates are represented as (X_r, Y_r, Z_r). In section 5.1 it explains that the receiver initial coordinate is fixed at (500, -5000, 6378000), this position is somewhere on the north pole as the Z-axis coordinate is permanent at 6378000 both for the receiver and the target.

Both target coordinates are (300, 0, 6378000) and (-300, 0, 6378000) respectively. The target distance from the receiver is 5km. the delay in the signal direct path is calculated by using the following formula.

Delay(direct) = D_{sr}/c seconds (5.2)

Where c is the velocity of the light and is equal to 299,792,458m/s

The distance for the indirect path is calculated as it is the sum of the distance from the satellite to target and target to the receiver which is in form of the equation is given by.

$D_{reflected} = D_{st} + D_{tr}$ (5.3)

The delay for each target is calculated by using the following formula incorporating suitable MATLAB commands.

Delay(indirect) = (D_{st}+D_{tr})/c seconds (5.4)

Table 5. 2

BeiDou Code Shifts-Up

Satellite No.	Tapping selection	Code shift
1	$1 \oplus 3$	712

2	1⊕4	1581
3	1⊕5	1414
4	1⊕6	1550
5	1⊕8	581
6	1⊕9	771
7	1⊕10	1311
8	1⊕11	1043
9	2⊕7	1549
10	3⊕4	359
11	3⊕5	710
12	3⊕6	1579
13	3⊕8	1548
14	3⊕9	1103
15	3⊕10	579
16	3⊕11	769
17	4⊕5	358
18	4⊕6	709
19	4⊕8	1411
20	4⊕9	1547
21	4⊕10	1102
22	4⊕11	578
23	5⊕6	357
24	5⊕8	1577
25	5⊕9	1410
26	5⊕10	1546
27	5⊕11	1101
29	6⊕8	707
30	6⊕9	1576
31	6⊕10	1409
32	6⊕11	1545
33	8⊕9	354
34	8⊕10	705

5.5 Determination of code offset

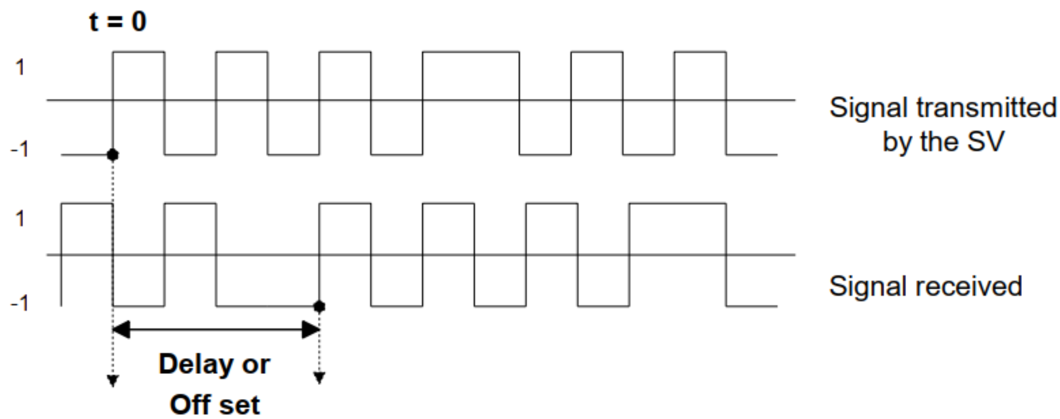

Fig 5. 3 Code Offset

As mentioned earlier that the code sequence for all satellite is generated by using MATLAB commands, these sequences (one allocated to each satellite) start at t=0. But the signal is delayed in practical scenario b/c when the signal is traveling from space it takes some definite amount of time to reach the receiver and thus delayed. We can use satellite transmitted signals as a reference to simulate the signal delay and thus do not need to calculate the total delay but only measure the difference in delay. In MATLAB, a function is written which calculates the difference in delay and thus it gives the delayed version of the PRN. In function PRN sequences of all the satellites are utilized, it utilizes the chip rate and difference of time of weak and delay calculated in section 5.1.3 above as input. This delay is called offset and the PRN is generated is called offset PRN. This function written in MATLAB is utilized to calculate offset for direct and as well as indirect signals. This offset is used to determine the baseband in-phase(I) and quadrature-phase (Q) components of the GPS signal as explained in the next section.

5.6 Extraction of the Baseband I and Q components

From the IF (intermediate frequency) signal the Baseband I and Q component can be extracted after determining the code offset. The direct signal is multiplied with the sine and cosine of the frequency we can get I and Q component. Therefore.

$$I_{direct} = d(t) x \sin \omega_0 t \qquad (5.5)$$

$$Q_{direct} = d(t) x \cos \omega_0 t \qquad (5.6)$$

Where *ωo* is the frequency in radians/seconds and includes the doppler frequency shift d(t) the code after offset has incorporated in case of this simulation. It is obtained after multiplying the B1i frequency in radians to the negative of delay calculated in section 4.4

For the reflected signal similar calculations are performed, phase-aligned to the direct signal first. As both the signal cover the different distances angle b/w direct and reflected signal is called phase difference as they are subjected to the dissimilar path, the phase is calculated by the following equation.

$$\varphi = \omega x \quad \text{path difference /c radians} \qquad (5.7)$$

Here the I and Q component of the reflected signals are given by

$$I_{reflected} = \alpha d(t) x \sin(\omega_0 t + \varphi) \qquad (5.8)$$

$$Q_{reflected} = \alpha d(t) x \cos(\omega_0 t + \varphi) \quad (5.9)$$

As usual ωo is the frequency in radians/seconds and includes the doppler frequency shift d(t) is the code after offset has incorporated. It is obtained after multiplying the B1i frequency in radians to the negative of delay calculated in 4.4 and α is the attenuation and will discuss in the next section.

To simplify the analysis, we assume that the receiver has some mechanism to lock on to the correct phase of the indirect signal hence ϕ is assumed to be zero.

The signal coming from every satellite is subjected to a doppler shift of the carrier frequency in proportion to the doppler frequency. Fortunately, this is taken care of inherently during the simulation by the sampling process. The doppler frequency shift is canceled out at each epoch during sampling.

5.7 Attenuation and noise

The signal transmitted by the BeiDou satellite is susceptible to noise and attenuation. In the section above it is mention that thermal noise has a significant effect on the signal. It's assumed that noise has a bandwidth of 2MHz and a noise temperature of 300^{o} K. Added the thermal noise to the receiver's noise Fig total noise power N is calculated its about-1411.40dBw. The BeiDou B1i PRN code power level is expressed in dBw received from SV .

5.8 Image formation

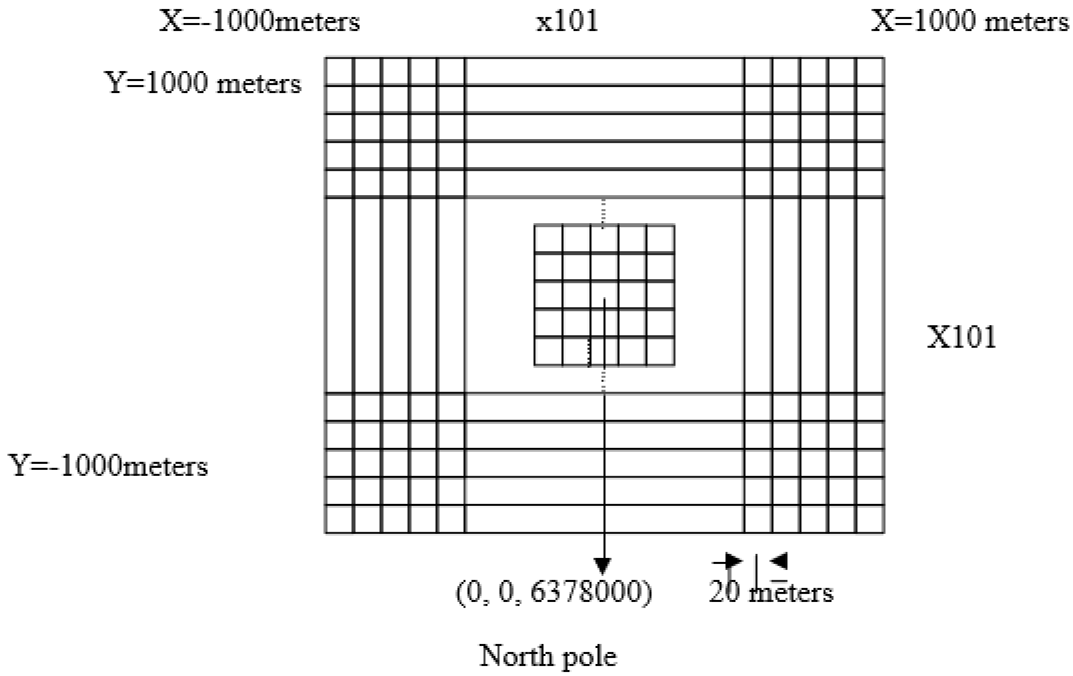

Fig 4. 5 Candidate position for target

The search area is initialized in which the target can be located as shown in Fig 44. To simulate the initialized area to simulate, the pixel size is 20 and two array of size 101 s10m. It's assumed that target always an area of 10m2. The target position is anywhere in the above matrix. The search area position is fixed at (0, 0, 6378000) and it is the search area center.

5.9 Reconstruction of image and processing of matched filter

In GNSS microwave remote sensing simulation the match filter process is the main part. As explained over in section 1, the linear FM auto-correlation function exhibits a narrow pulse property, match filter processing it passes through the filter the output will become narrower. This in term improves the system resolution. Two codes multiplied bit by bit and add them up in MATLAB. The process is carried out for every epoch or sample and relevant values are stored in the 101x101 matrix as shown above. In this simulation two signals are used for matched filter processing the input signal that consists of the direct and reflected signal coming from all satellite in view and the test signal that is received only from the reference satellite.

For the test satellite first, we must calculate the delay and offset PRN. By calculating the distance between reference satellite to candidate position and receiver to candidate position we can determine these parameters. The delay and offset PRN are determined in the same way as described in sections 4.4 and 4.5, respectively. We can get only the PRN sequence of the reference SV and calculate the delay, not for only the candidate position but all the candidate expected position in the area of interest so that the distance is calculated not for the only target fixed position but all target expected positions. As result, the test signal is collapsed to baseband by multiplying the instantaneous value of the by the sine of B1 frequency in radian to the negative of the delay calculated above.

Each epoch or sample of both signals are multiplied and added, and a different result of the correlation is obtained for each instant of time. These values are stored in the matrix with suitable commands. The matrix is scanned for each sample in time, rendering the procedure equivalent to the matched filter process. It multiplied the signal and its delayed version and adds them for all epochs, this is like integrating over the sampling period. At the location of the target had a maximum value or correlation. It is apprized that the values for the baseband in-phase (I) and quadrature-phase (Q) components are converted to equivalent magnitude by the following simple equation.

$$\text{Equivalent Magnitude} = \sqrt{I^2 + Q^2} \qquad (5.10)$$

By using the above-mentioned information a grayscale image is formed. This image is constituted of colors that are a mixture of black, white, and shades of gray. In the above mention simulation, the MATLAB function returns to the maximum value, the data is inverted and normalized so that the maximum value is one. When the correlation is maximum this is achieved, and data is displayed as black in the pixel or square and when there is no target there is minimum correlation then the pixel or square is displayed as white. Thus, the image of the target is formed directly pinpoints at the desired coordinate in the candidate position. The MATLAB function IMAGE is used to culminate in the task of image generation.

6. Result's and analysis

The simulation was carried out on MATLAB with different conditions and Scenario by changing the parameter's as shown below: -

Table 6. 1 Parameters for Simulation

Sr. No	Description	Values
1.	Integration time (Ts)	0.1 Seconds
2.	Attenuation Factor	0.5
3.	Noise	0
4.	Target No.1 coordinates	[200, 0, 6378000]
5.	Target No.2 coordinates	[-200, 0, 6378000]
6.	Receiver coordinates	[500, -500, 6378000]
7.	Receiver velocity	[300, 0, 0,]
8.	SV Involves	[1, 7, 9, 14, 16, 22, 25, 26, 27]
9.	Reference SV	6

The Integration time is the period over which the signals are added, whereas attenuation factor is the reflected signal's attenuation as compared to direct signal. The Receiver and targets coordinates have been selected on north pole to simplify the simulation.

6.1 Case No.1 with integration time 0.1seconds and no noise

The simulation was performed with the parameter's values given in Table 6.1 above which resulted in the generation of Fig 6.1. In which two targets can easily be recognized with good spatial resolution. The simulation was performed with an integration time 0.1 second and thus consumed lesslow time but the resolution of the image is affected and after the introduction of noise the target will no longer be resolved.

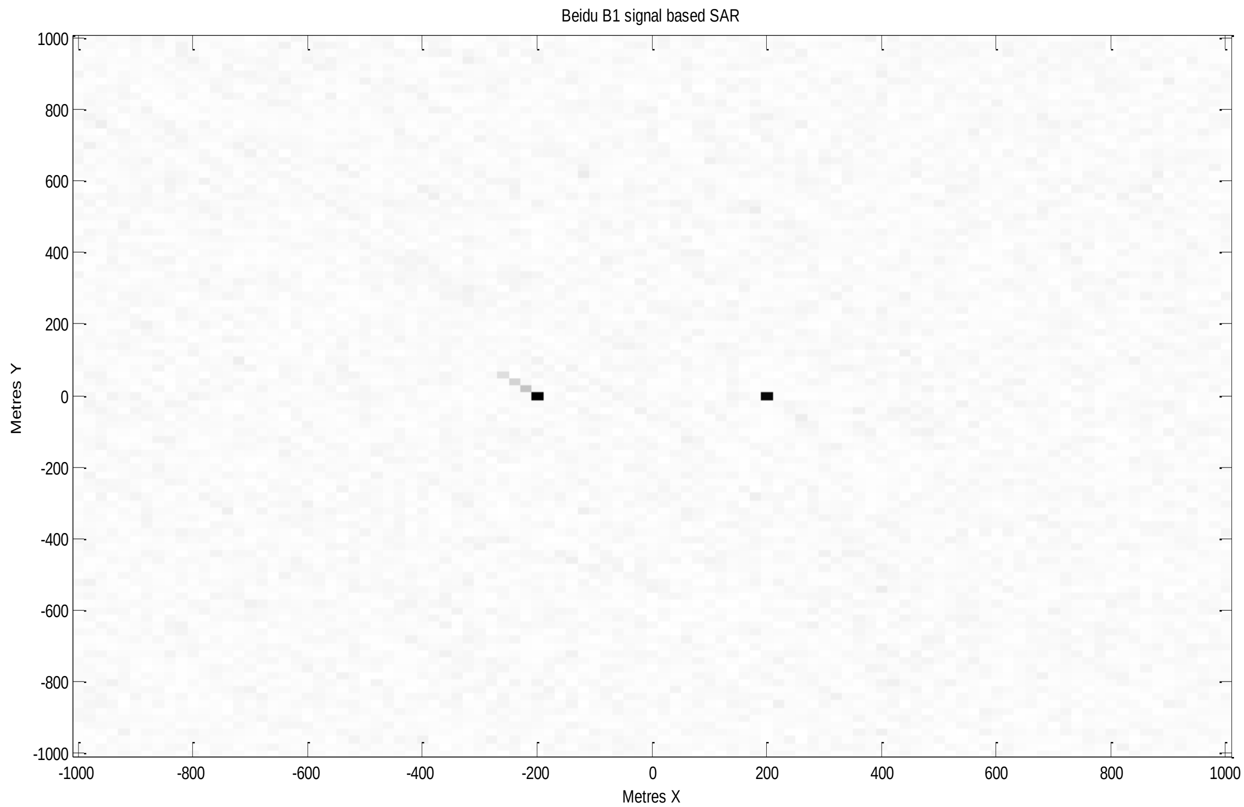

Fig 6. 1 Integration Time 0.1 Seconds and Without Noise

6.2 Case No.2 with integration time 0.1 seconds and noise 16dB

The noise was added while other's parameter values remain the same as given in Table 6.1. From Fig 6.2 the target is still recognized but the noise has been introduced in the background.

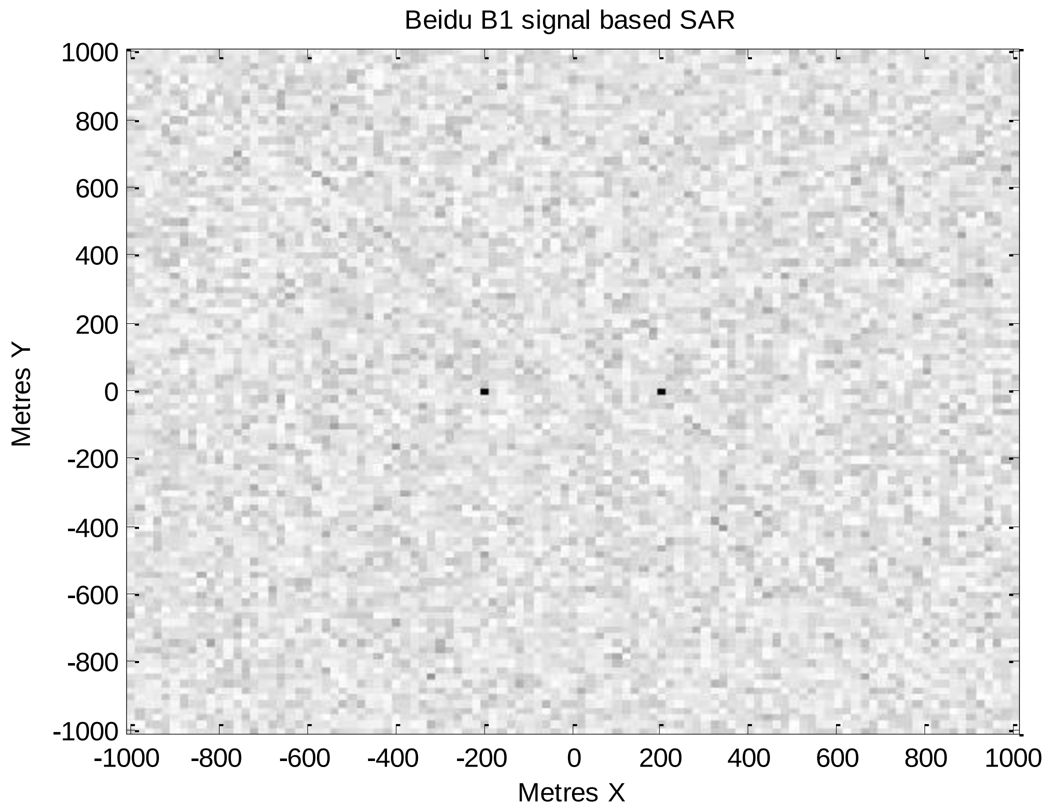

Fig 6. 2 With Integration Time 0.1 Second and Noise 16dB

6.3 Case No.3 with integration time 0.2 seconds and noise 16dB

The simulation integration time is increased to 0.2 seconds, though the simulation took a longer time to execute but improved the result and resolution. It is kept in mind while choosing integration time it's not very small or too long. For very small coherent integration time will not allow to build up the matched filter output to it is full potential as well for too long will not improve over an increasing summation of independent samples.

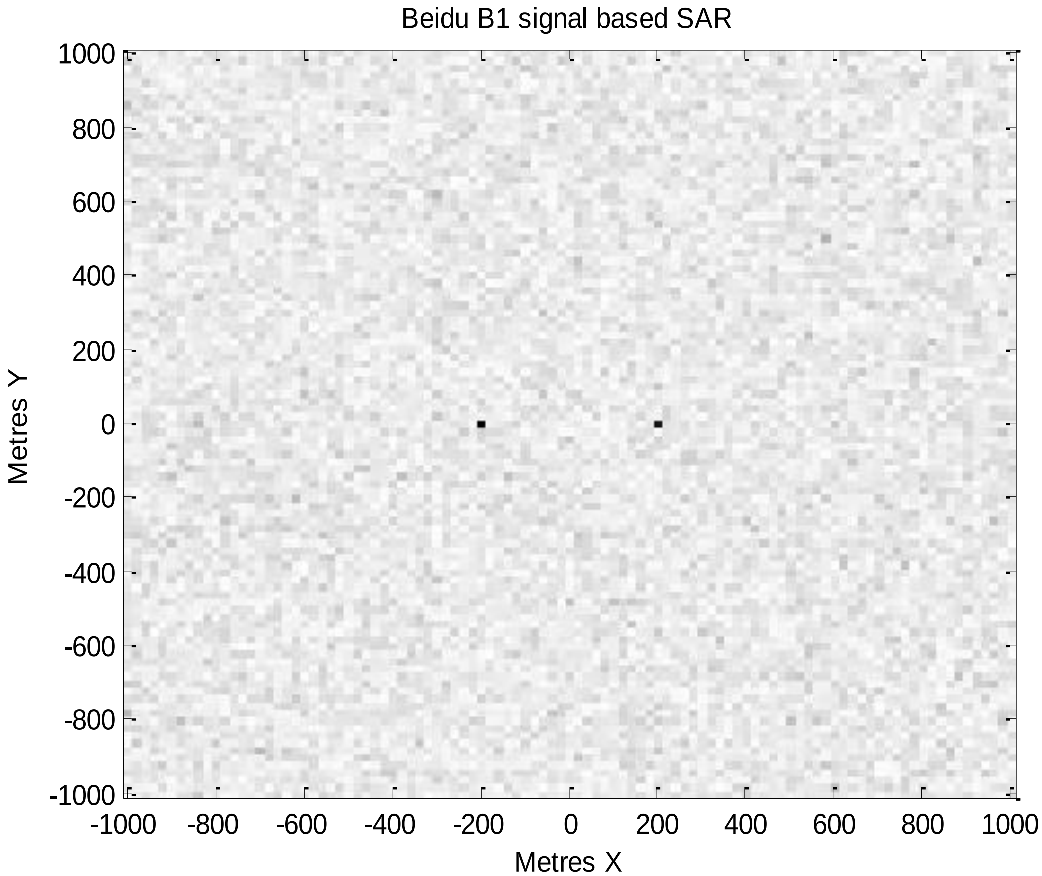

Fig 6. 3 Integration Time 0.2 Second and Same Value of Noise

6.4 Case No.4 with integration time 0.1 seconds and target moving with different velocity.

In this case, the simulation starts, and time is kept at 0.1 seconds and no noise was introduced. Target moves at different velocity to check the result in the first case target move at the speed of 36km /hr (10m/s) away from each other. The target moving away, from the depict Fig 6.4 confirmed the simulation. In the next case targets moving away with the speed of 72km/hr (20m/s) in Fig 6.5 simulation showed the results. However, when the speed increase more to 100m/s (362km/hr) then from the simulation results it clear that the target no longer can be resolved simulation results shown in Fig 6.6. From the simulation results its concluded that only static targets, ships, or slow-moving terrestrial targets can be effectively be recognized with this technique.

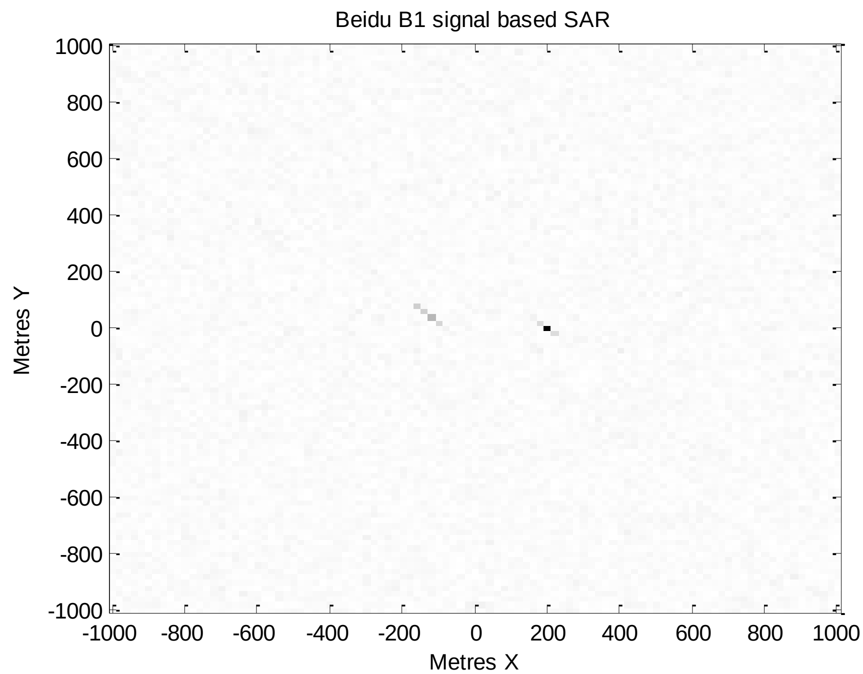

Fig 6. 4 Integration Time 0.1 Seconds and Target Moving With 10m/s

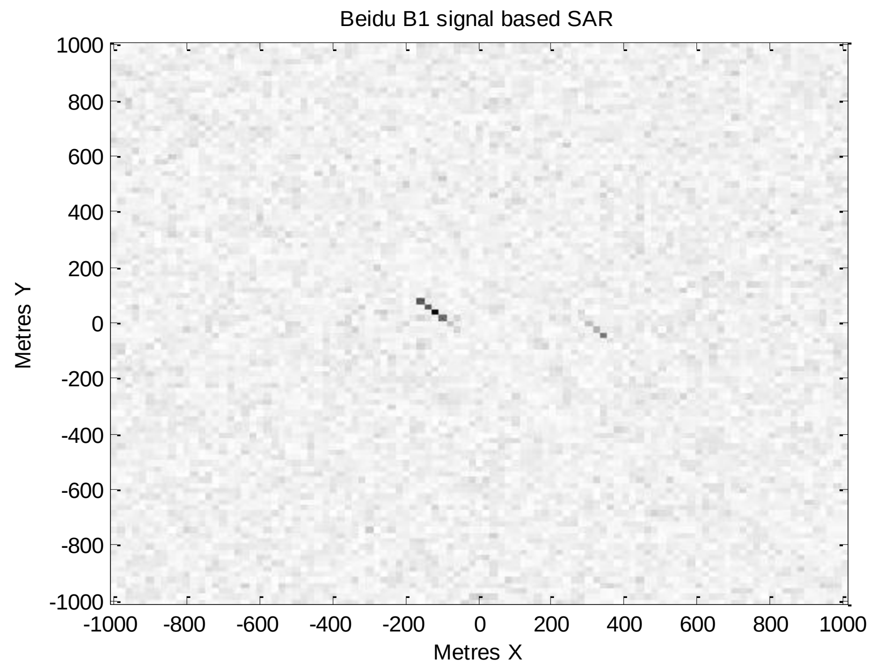

Fig 6. 5 Integration Time 0.1 Seconds and Target Moving With 20m/s

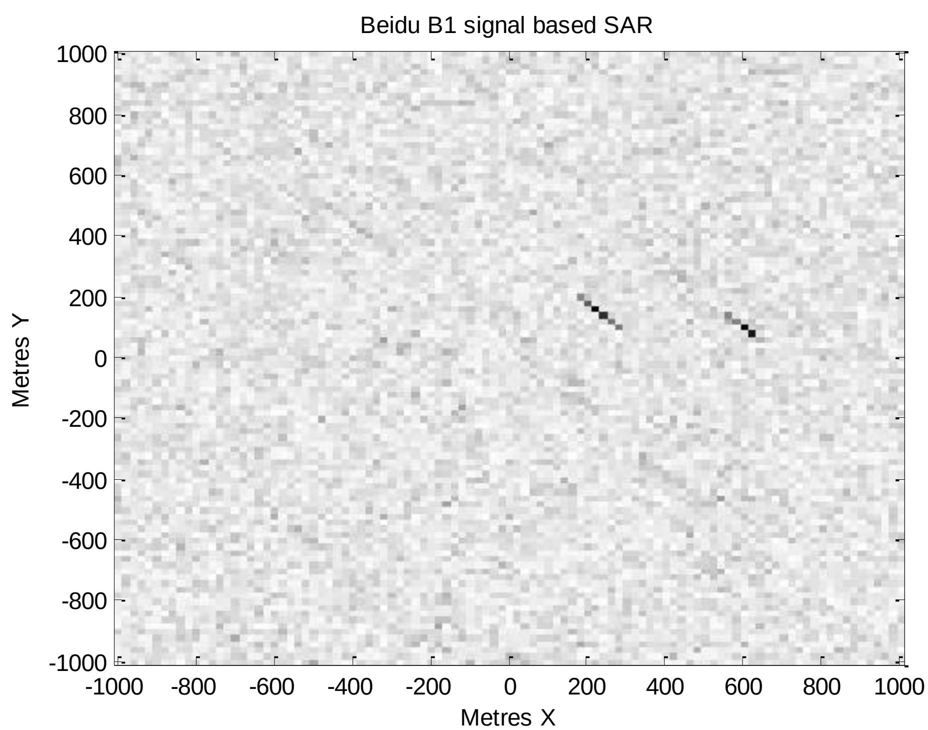

Fig 6. 6 Integration time 0.1 Seconds and Target Moving with 50m/s

6.5 Case No.5 with integration time 0.1 seconds and receiver moving with different speed

In this case, no noise was added, the integration time is kept the same at 0.1 Seconds. While the receiver velocity is increased to [500, 0, 0] m/sec, Fig 5.5 depicts the simulation result it concludes that the greater the speed of the receiver, the higher the resolution. This result is in line with the SAR principle.

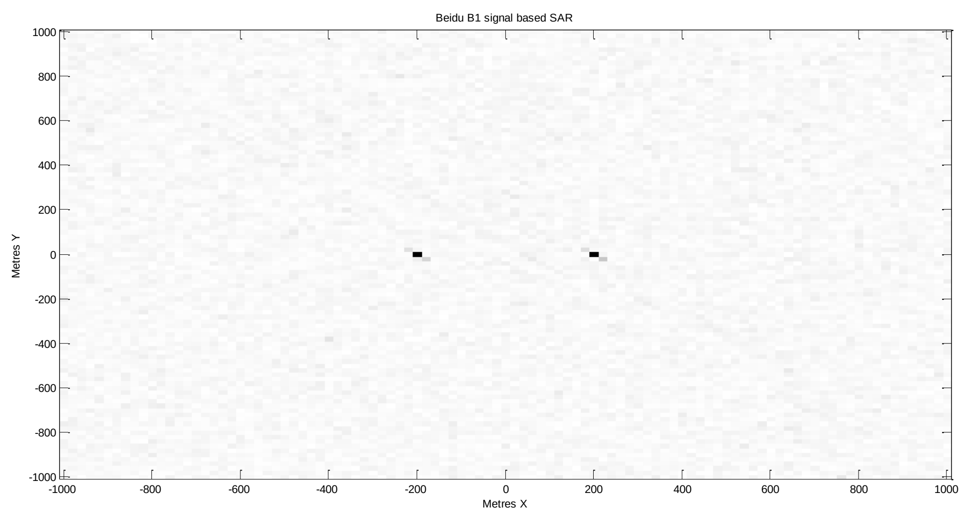

Fig 6. 7 Integration time 0.1 second and Receiver Moving with Different Speed

6.6 Case No.6 Attenuation of reflected signal

In this case, attenuation is changed to 0.3 from 0.5 (as in case 1). Which resulted in low power of the reflected signal, but targets are still recognizable though some noise is observed in the background evident from Fig.

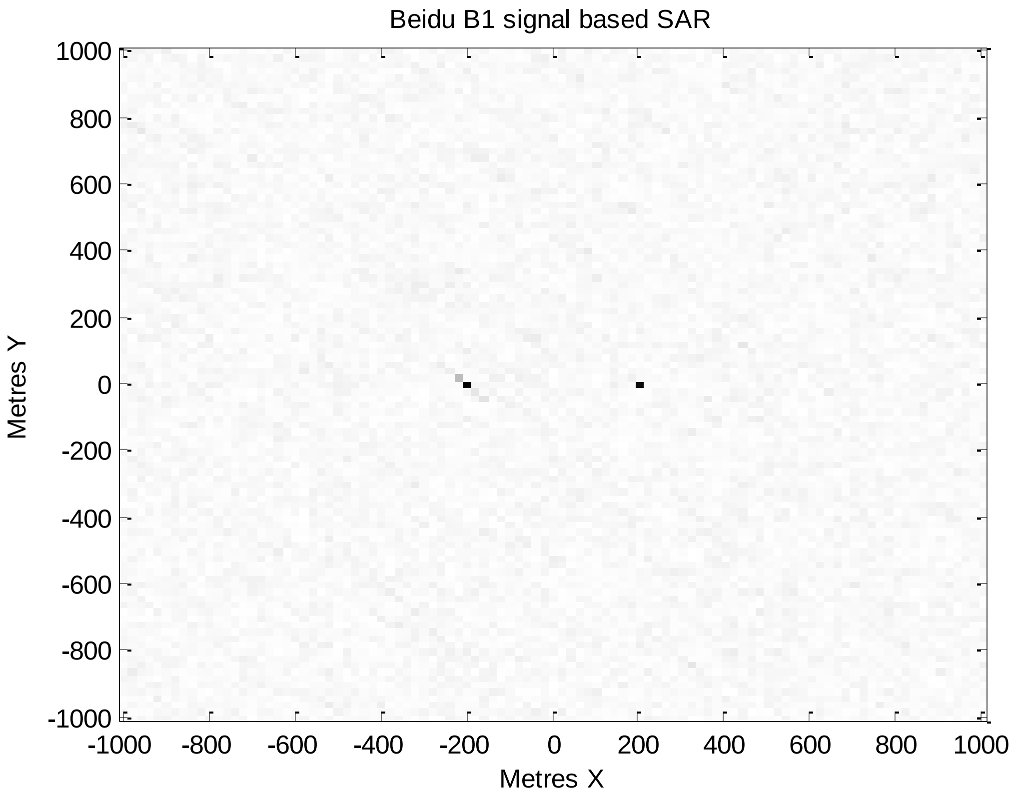

Fig 6. 8 Attenuation of 0.3 for Reflected Signal

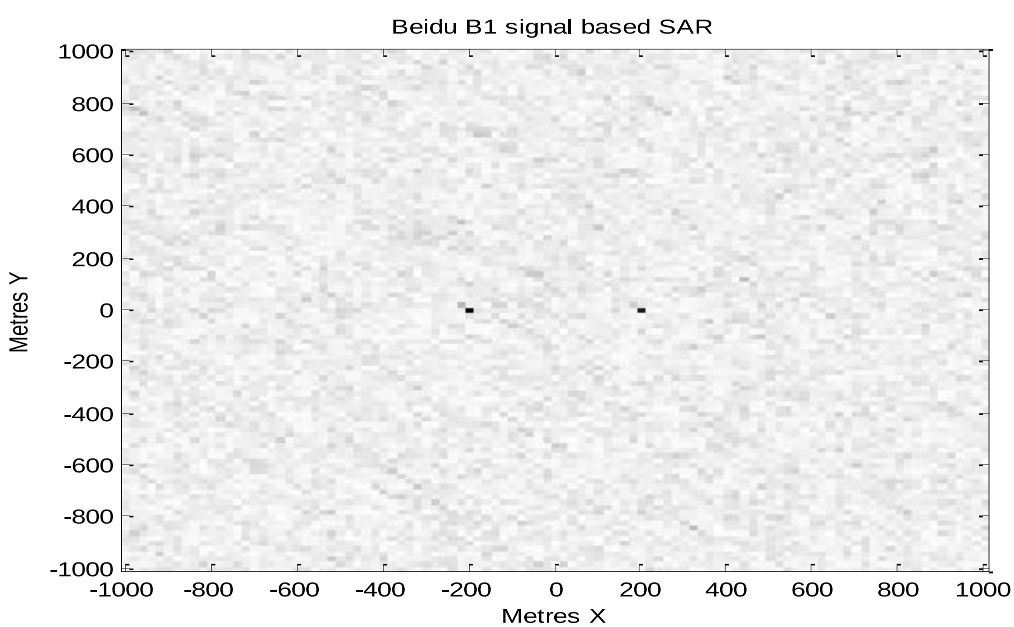

Fig 6. 9 Attenuation of 0.1 for Reflected signal

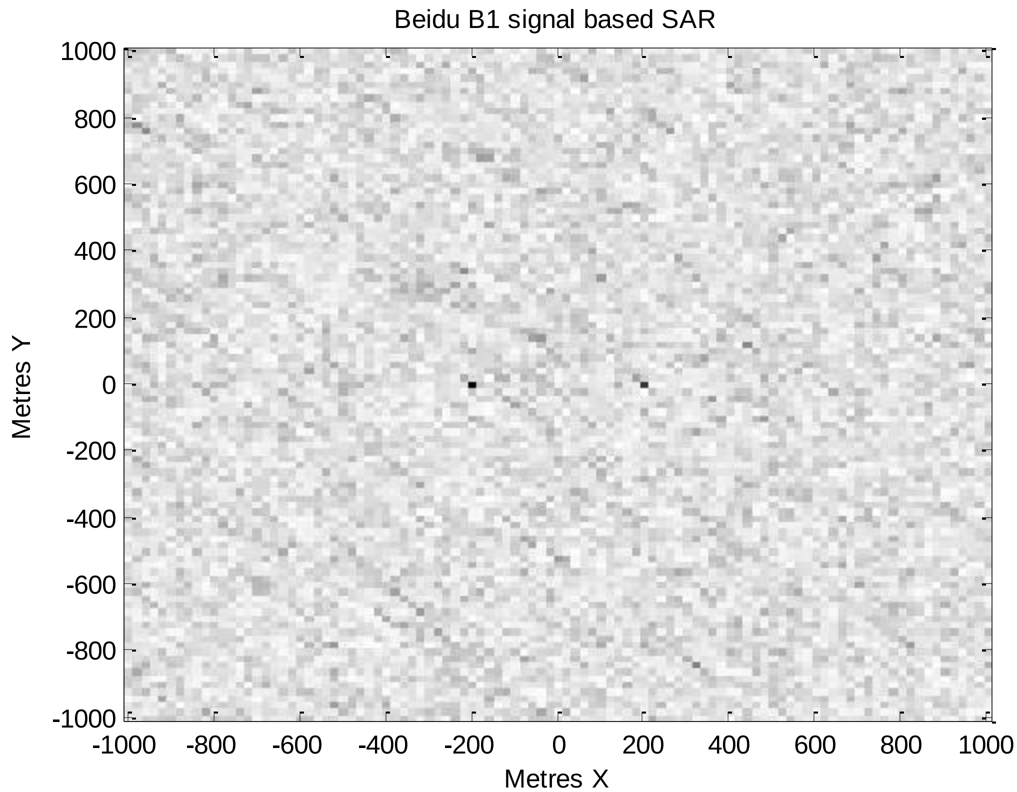

Fig 6. 10 Attenuation of 0.05 for reflected signal

7. Conclusion and recommendations

The objective of the project was demonstrating the GNSS based passive SAR concept using BeiDou signals as Transmitter of Opportunity. The project objective has been achieved and the imaging system for reflected BeiDou system signals with a matched filter has successfully been simulated in MATLAB version R2014a.

Under various condition code was tested, the various practical scenario was simulated in line with a project feasibility report. The receiver can move at different velocity and the effect of noise attenuation and interference on both direct and reflected signals was analyzed carefully and in a detailed manner. The BeiDou modified receiver which may be utilized to materialize the simulation was discussed and in this respect block diagram has been generated.

During the simulation process, many interesting results have been observed and stated as follow:-

(a) The number of samples has been directly proportional to the time of observation. Increasing the time of observation will increase the number of samples and in result better simulation results and improve resolution by canceling the noise.

(b) When the movement in the target is introduced then it is recognized from the simulation results that only static targets, ships, or slow-moving terrestrial targets will be effectively recognized with this technique.

(c) During the simulation, it has been recognized that the greater the receiver speed higher the resolution. These results are in line with the SAR principle.

(d) Unexpectedly, during the simulation in the above cases when the integration time was increased in the presence of noise the target will not further be resolved . that reminded us of low power of reflected signal and in presence of noise extraction of useful data remains the biggest challenge in the remote sensing environment under the contemporary circumstances.

Due to a shortage of time, simulation of certain scenarios could not be explore in detail. In terms of time and effort, there are many features in the project that need exhaustive input and can be tackle in the future undertaking. These are summarized in the next paragraphs.

In a more detailed manner, the proposed modified receiver can be studied. If possible, a working "proof of concept" development model of the system can be assembled and tested under a practical situation.

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